

User Manual

TDS Toolkit

File, Waveform, Mask and Limit Test Software for TDS 500, TDS 600
and TDS 700 Series Digitizing Oscilloscopes

Version 1.0.0

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TDSTK-100

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Figures 1–1, 1–2, 1–3 and 8–3 are reproduced from Tektronix manuals 070-9876-00 and 071-0627-02 for the purpose of identifying the connectors and controls the reader is asked to use.

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General Safety Summary

Review the following safety precautions to avoid injury and prevent damage to the oscilloscope or to any product connected to it. To avoid potential hazards, use the oscilloscope only as specified in its own manuals, and use this program only as specified here.

Only qualified personnel should perform service procedures. Reaching the memory protection switch described in Section 8 requires access to the oscilloscope's cabinet.

To Avoid Fire or Personal Injury

Ground the Product. The oscilloscope is grounded through the grounding conductor of its power cord, and the GPIB cable screen bonds its chassis to the controller. To avoid electric shock, both instruments must be connected to earth ground before any signal connection is made.

Connect and Disconnect Properly. Do not connect or disconnect the GPIB cable while either instrument is switched on where the two are fed from different supplies. Connect the controller's cable to the oscilloscope before applying power to the signal source.

Do Not Operate Without Covers. Do not operate the oscilloscope with covers or panels removed. The memory protection switch is reached from outside the cabinet; nothing in this manual requires the cabinet to be opened with power applied.

Observe All Terminal Ratings. This program can switch channels on and change vertical scale factors over the bus. It cannot know what is connected to the inputs. Observe the ratings marked on the oscilloscope.

Do Not Operate With Suspected Failures. If the oscilloscope reports diagnostic failures on the System tab, have it inspected by qualified service personnel before relying on any measurement it makes.

Data Loss Summary

This program writes to the oscilloscope. The following actions change or destroy data on the instrument and cannot be undone from this program. Each is described in full where it appears.

Irreversible Actions

Delete. Deleting a file or a folder on the instrument is final. There is no recycle bin on the instrument, and removing a folder removes everything inside it. See *Deleting Files and Folders* in Section 2.

Upload. Writing a file replaces any file of the same name without a further question, because the program sets FILESYSTEM:OVERWRITE ON when it connects. See *Uploading Files* in Section 2.

Send to Instrument. Writing a waveform into a reference memory replaces whatever that reference held. See *Sending a Waveform to the Instrument* in Section 4.

Create Template and Use This Envelope. Both write a limit template into a reference memory, replacing its contents. See Section 5.

Clear Errors. Clearing the instrument's error log erases its error history, which is kept in non-volatile memory. See Section 7.

Secure Erase. A secure erase writes zeroes over every waveform and setup memory. Calibration data is left untouched. See Section 8.

Recall Factory Setup. This returns the instrument to its default settings. The GPIB address, the calibration constants and protected user data are left untouched. See Section 8.

Options. Writing the factory option words writes directly to non-volatile memory. Nothing can be read back to confirm what was written. See Section 8.

Preface

This is the User Manual for TDS Toolkit, a program that operates a Tektronix TDS 500, TDS 600 or TDS 700 series digitizing oscilloscope from a Windows computer over the General Purpose Interface Bus. It describes what each of the program's eight tabs does, what each control sends to the instrument, and what the instrument does in reply.

The program is written for the instruments that carry the filesystem, hardcopy, mask and limit-test subsystems described in the TDS Family Programmer Manual. Which of those an individual instrument has depends on its firmware. The program establishes that when it connects and offers only what the instrument can do; Appendix A lists what was found in each firmware image.

The firmware of the following models was read directly, and what each version can do is listed in Appendix A. Other instruments in the same families work as well; they are asked when they connect rather than looked up.

TDS 500: TDS 520, TDS 520A, TDS 520B, TDS 520C, TDS 524A, TDS 540, TDS 540A, TDS 540B, TDS 540C, TDS 544A, TDS 580C

TDS 600: TDS 620, TDS 620A, TDS 620B, TDS 640A, TDS 644A, TDS 644B, TDS 680B, TDS 684B, TDS 684C

TDS 700: TDS 714L, TDS 744A, TDS 754C, TDS 754D, TDS 784C, TDS 784D, TDS 794D

This manual assumes you can operate the oscilloscope from its front panel and that its GPIB interface is configured. It does not repeat the instrument's own documentation.

Related Manuals

The following Tektronix documents describe the instrument itself and the commands this program sends:

■ *TDS Family Digitizing Oscilloscopes Programmer Manual*, 070-9876-00, describes every command quoted in this manual.

■ *TDS 500D, TDS 600C, TDS 700D & TDS 714L User Manual*, 071-0130-03, describes operating the oscilloscope from its front panel.

■ *TDS 500D, TDS 600C, TDS 700D & TDS 714L Service Manual*, 071-0627-02, gives the location of the memory protection switch used in Section 8.

■ *TDS 500D, TDS 600C, TDS 700D & TDS 714L Performance Verification and Specifications*, 071-0630-03, gives the specifications the instrument is warranted to.

Getting Started

Product Description

What the Program Does

TDS Toolkit operates a TDS 500, TDS 600 or TDS 700 series digitizing oscilloscope from a Windows computer over GPIB. The instruments carry a filesystem, a hardcopy subsystem, a mask subsystem and a limit-test subsystem, all of which are reachable over the bus and none of which the instrument itself provides a way to work with from a computer.

The program is arranged as eight tabs. Each is described in its own section of this manual:

- **Files** browses the instrument's drives and moves files either way. See Section 2.
- **Screenshot** captures the instrument's display through its hardcopy port. See Section 3.
- **Waveforms** captures live channels and stored references, plots them, and writes them back. See Section 4.
- **Limits** builds a limit template and runs a limit test against it. See Section 5.
- **Masks** draws, sends and judges masks, and sets the instrument up for an eye diagram. See Section 6.
- **Error Log** reads the instrument's own error history. See Section 7.
- **System** reads and sets the instrument's identity, clock, ports, diagnostics and factory options. See Section 8.
- **Settings** holds the plot colours and the saved-picture resolution. See Section 9.

Nothing is sent to the instrument that the instrument's own Programmer Manual does not describe, with one exception: the factory option words in Section 8 are in no Tektronix manual and were established on the bench.

What Your Instrument Can Do

The firmware decides what is possible, and the range spans nine years of it. Three generations behave differently:

Table 1–1: What each firmware generation offers

Firmware	Browse	Download	Upload
v2.x (TDS 520, 540, 620)	no filesystem	no	no
A and B series, v3.x to v4.x	yes	yes	no
C and D series, v5.0e and later	yes	yes	yes

The A and B series have no FILESYSTEM:WRITEFILE command, so no program can send a file to them over GPIB. They browse and download normally. Waveform transfer works on every instrument in the range, including those with no filesystem at all.

NOTE. The program establishes what the instrument can do when it connects and greys out what that firmware cannot do, rather than letting you start something that was never going to finish. Appendix A gives the full table.

Requirements

Table 1–2: Equipment required

Item	Minimum requirement
Oscilloscope	TDS 500, TDS 600 or TDS 700 series with the GPIB interface
GPIB controller	Any controller with a VISA driver, for example a National Instruments GPIB-USB-HS or a Keysight 82357B
GPIB cable	IEEE Std 488.1-1987, Tektronix part number 012-0991-00
Computer	Windows, with a VISA runtime installed

VISA Runtime

A VISA runtime is a system driver, not a part of this program, and nothing can reach the instrument without one. Install either NI-VISA or the Keysight IO Libraries Suite, and install the GPIB controller's own driver first, because VISA discovers the hardware through it.



CAUTION. Install one VISA runtime, not two. Two implementations on one computer compete for the same interface and the symptom is an instrument that answers intermittently or not at all.

Connecting the Instrument

The GPIB Connector

The oscilloscope has a 24-pin GPIB connector on its rear panel. The connector has a D-type shell and conforms to IEEE Std 488.1-1987. See Figure 1–1.

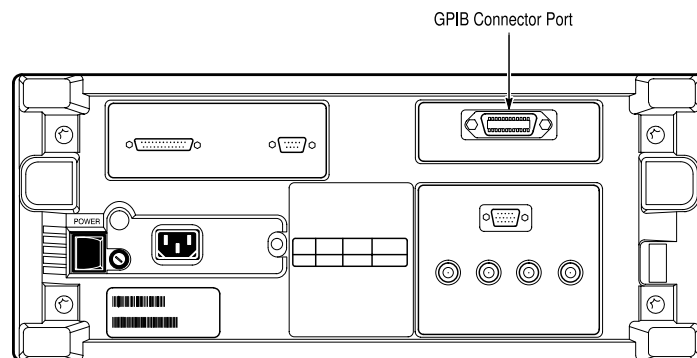


Figure 1-1: GPIB connector location on the rear panel

Attach the GPIB cable to this connector. Where more than one instrument is on the bus, the connectors stack: each cable's plug carries a socket on its back so the next cable can be attached to it. See Figure 1-2.

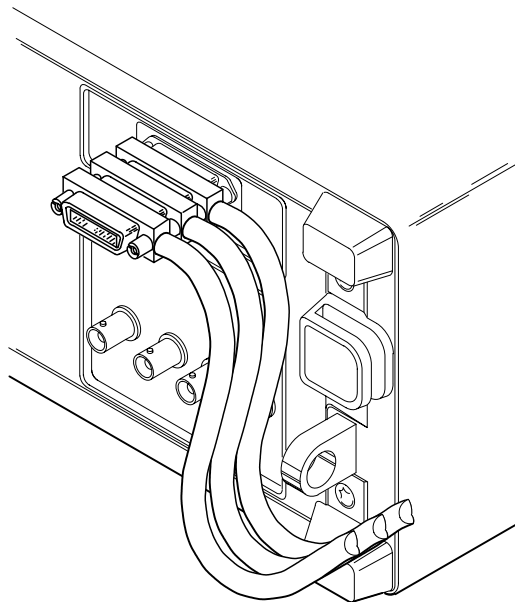


Figure 1-2: Stacking GPIB connectors

Observe the bus rules the instrument's own manual gives: no more than 15 devices on one bus, no more than 20 metres of cable in total and no more than 2 metres between devices, and at least two-thirds of the devices powered on. Devices may be connected in a star, in a line, or in any combination of the two. See Figure 1-3.

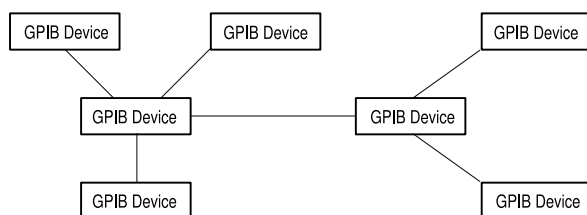


Figure 1-3: Typical GPIB network configurations

Setting the Address

Set the instrument's GPIB address from its front panel, and note what it is. On a TDS 500, 600 or 700 series instrument, press **SHIFT**; then press **UTILITY**. Press the main-menu button **System** until **I/O** is highlighted in the pop-up menu, press the main-menu button **Port**, and select **GPIB**. The address is set from the same menu under **Configure**.

NOTE. The instrument must be in Talk/Listen, not Off Bus or Hardcopy (Talk Only). In Talk Only it will not answer a query, and this program's scan will not find it.

Installing the Program

From the Executable

Copy `TDS Toolkit.exe` to a folder you can write to and run it. The program is self-contained: it needs no Python, installs nothing and writes nothing to the registry.

Three things are kept in the folder the executable is in, and are made there the first time it runs:

tdstoolkit.json	The settings: the language, the remembered instrument address, the plot colours, any colour schemes you have saved, and the saved-picture resolution.
tdstoolkit.log	What the program did, and anything the instrument reported that was not expected. Rotated to <code>tdstoolkit.log.old</code> at 512 kbytes.
masks	The mask library. The masks supplied with the program are copied here the first time it runs, and anything you draw and save joins them.

From Source

The program needs Python 3.8 or newer with Tkinter, which the Windows installer includes. Install the two packages it uses and run it:

```
pip install -r requirements.txt
python "TDS Toolkit.pyw"
```

`pyvisa` is the Python binding for the VISA runtime and is required. `tkinterdnd2` is optional and adds dragging files in from Explorer; without it everything else works and the title bar says so.

Starting the Program

The First Connection

Nothing needs configuring. The program tries GPIB0::3::INSTR, and if nothing answers it scans the bus and opens the instrument picker. Every address VISA knows about is asked to identify itself with *IDN?, which changes nothing on any of them, and anything answering as a Tektronix TDS is marked as an oscilloscope.

1. *Choose the instrument:* Click its row in the picker, or type an address into the **Address** box.
2. *Remember it:* Leave **Remember this address** ticked. The address is written to the settings file and the next run connects straight to it.
3. *Connect:* Press **Connect**, or double-click the row.

The instrument can be changed at any time from the **Instrument** dropdown above the tabs, which lists every TDS oscilloscope the last scan found. Its last entry, **Other address...**, opens the picker again. **Scan** beside it sweeps the bus; pressing it a second time while it runs cancels it.

NOTE. VISA lists every address it has ever been configured with, so a bus with one live instrument can show half a dozen. The picker gives the reason each address gave, so an instrument that is switched off can be told from one that was never there.

The Window

The window is the same on every tab: the instrument dropdown, the **Scan** button and the language button along the top; the eight tabs below them; and a status line along the bottom, with a progress bar at its right while work is in hand. See Figure 1–4.

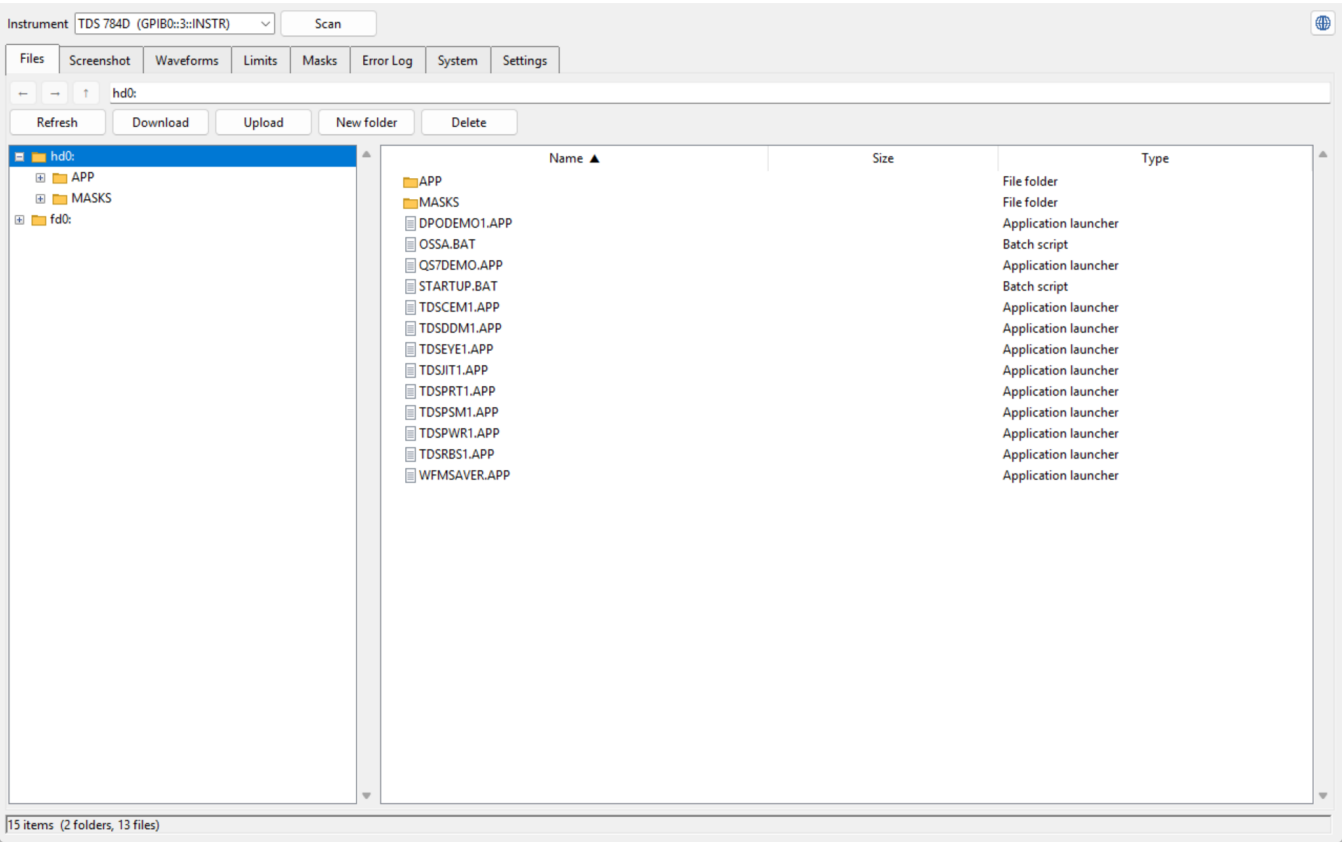


Figure 1-4: The program window, showing the Files tab

While the Program Is Busy

The status line is where the program answers. Every action puts a line there saying what it is doing, and another saying what happened. Anything unexpected that the instrument reports is shown there as well and written to the log.

One operation runs at a time. While the program is talking to the instrument the pointer becomes a watch and the buttons that would start another operation are greyed, on every tab, so that a transfer on one tab cannot be started on top of one on another. The drawing tools on the Masks and Limits tabs are not greyed: a transfer to an instrument is no reason not to be able to draw.

A button greyed for the whole session, rather than for the length of an operation, is one this firmware has no command for. The reason is given once, when the instrument connects.

Conventions

Throughout this manual the following conventions apply:

- The name of a control in the program is set in **bold**, exactly as the program labels it in English: **Get waveform**, **Send to instrument...**
- The name of a control on the oscilloscope is set in bold and in capitals as it is engraved on the panel: **SHIFT**, **UTILITY**, **CLEAR MENU**.
- Commands sent over the bus are set in **Courier**, in the form the program sends them: **FILESYSTEM:READFILE**.
- A message the program shows is quoted in double quotes, word for word in English.
- Events the instrument raises are given by number and by the instrument's own wording: event 2241, "waveform requested is invalid".

Measurements quoted as measured were taken on a TDS 784D running firmware v7.4e, a TDS 784C running v5.3e or a TDS 640A running v3.8.8e, and the instrument is named where it matters.

Command Line

The program takes the following switches. An address given here overrides the remembered one for that run only and is not written to the settings file.

Table 1-3: Command line switches

Switch	Effect
--address=GPIB0::1::INSTR	Connect to this address instead of the remembered one. --address GPIB0::1::INSTR, with a space, is also accepted.
--version	Print the version and exit.
--selftest	Connect, probe the volumes, list a folder, read a file, write and verify a test file, and remove it. No window is opened. Intended for checking the plumbing after a change of driver or cable.
--check-translations	Audit every language file beside the program and exit non-zero if any is wrong. No window is opened.

Files

The Files tab browses the instrument's drives and moves files either way. It is the tab the program opens on.

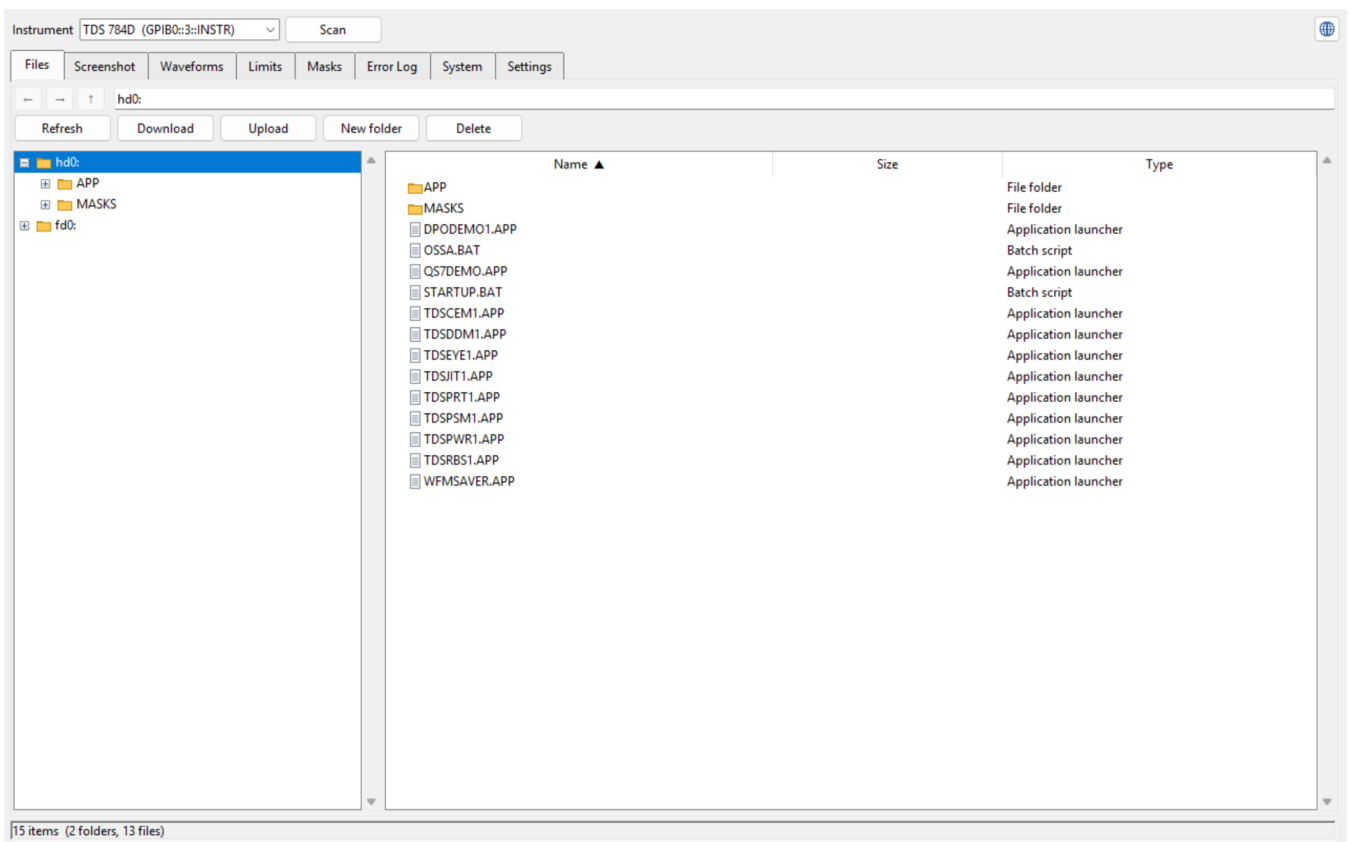


Figure 2–1: The Files tab

The Panes

Folder Tree

The left pane lists the drives the instrument reports and the folders on them. The program probes for hd0:, fd0:, hd1: and fd1: when it connects, because the instrument has no command that lists its own volumes, and opens hd0: if it is there and the first volume it found if it is not.

Click a folder to go into it. The plus in a box beside a folder opens it in place without leaving the folder you are in, and becomes a minus to close it again.

File List	The right pane lists what is in the current folder. It has three columns:
Name	The name as the instrument holds it, which is always in capitals.
Size	In bytes, with thousands separated. Normally empty: the instrument's directory listing carries no sizes, so a size is known only after that file has been transferred, and is filled in then.
Type	What the extension means. A folder reads "File folder"; an extension the program does not know reads " <i>EXT</i> File".

Click a column heading to sort by it, and click the same heading again to reverse it. Folders sort above files whichever column is sorted. Sorting is done from the listing already in hand; the instrument is not asked again.

NOTE. *Sizes are not in the instrument's directory listing and there is no command that returns one. The Size column is therefore filled in as files are transferred, and a folder you have only browsed shows none.*

What Is Not Listed

A card imaged from Windows carries VFAT long-filename records for any name Windows did not consider 8.3-clean. The instrument's FAT16 driver predates VFAT and reads each of those records as an ordinary directory entry, reporting it as a file with a two-character name and a trailing dot. Those entries are filtered out before the list is drawn.



CAUTION. *Do not attempt to delete such an entry by any other means. It is not a file: it is the record that holds the long name of the file beside it, and removing it strips the long name off that file.*

Moving About

The three arrows above the list are Back, Forward and Up, and the box beside them is the current path. Type a path into it and press Enter to go there.

Table 2–1: Navigation keys

Key	Action
Alt+Left	Back to the folder before this one
Alt+Right	Forward again
Alt+Up, Backspace	Up one folder
F5	Read the current folder from the instrument again
Double-click, Enter	Open a folder, or save a file
Ctrl+A	Select every row
Delete	Delete what is selected

Back and Forward follow the browser convention: arriving at the folder you came from counts as a step back rather than a new destination, and going somewhere new discards the trail ahead of you.

NOTE. *Backspace and Delete act only while the Files tab is in front. The arrow combinations and F5 are not gated that way.*

How Long a Listing Takes

The instrument's `FILESYSTEM:DIR?` returns names with nothing to say which are folders, so each entry is classified by trying to change directory into it: two or three queries a name. The root of a hard disk holding 105 entries takes about three seconds. Each folder is listed once and remembered; **Refresh** reads it again.

Downloading Files

Select what you want and press **Download**, or double-click a single file. What happens next depends on the selection:

- **One file.** A Save As dialog opens with the instrument's name filled in.
- **Several files.** A folder is asked for, and each file is written as it arrives. Anything already there of the same name is listed and confirmed first.
- **One folder.** A folder is asked for, and the whole tree below it is brought down into it. The tree is enumerated first, so the progress bar has a total to count against.

Folders are done one at a time. Selecting several and pressing **Download** raises "One folder at a time", because saving a folder brings down everything inside it.

The transfer itself is `FILESYSTEM:READFILE` where the firmware has it. Where it does not, the program falls back to `FILESYSTEM:PRINT <path>, GPIB`, which is how the A and B series hand over a file.



CAUTION. *The PRINT route is tried at most twice for any one file. A long run of failed prints puts the instrument's filesystem out of action until it is power cycled, and that is why the program stops rather than retrying.*

Uploading Files

Press **Upload** and choose a file, or drag files onto the list from Explorer. The file is checked against the instrument's 8.3 name rule, written, and read back.

1. *Choose the destination:* Go to the folder the file is to be written into. Uploads go to the folder the list is showing.
2. *Choose the file:* Press **Upload** and pick it, or drag it onto the file list.
3. *Correct the name if it is refused:* A name that does not fit the 8.3 rule is refused with the reason, and a box opens with the name in it to be corrected. Names dragged in from Explorer are shortened automatically instead.
4. *Confirm:* The program states the file, its size and the destination. CHECK that they are what you intended, then press Yes.

Every upload is verified. The target is deleted, the file is written, the directory is re-read to confirm the name appeared, and the file is read back and compared byte for byte. Up to four attempts are made; a failure is reported as a failure rather than guessed at.

Names on the Instrument

The instrument's filesystem is FAT16 and its names follow the DOS 8.3 rule: up to eight characters, then optionally a dot and up to three more. Letters, digits and the punctuation \$ % ' - _ @ ~ ` ! () { } ^ # & may be used. The name is stored in capitals whatever case it is given in.

NOTE. *Dragging files in from Explorer converts the names rather than refusing them: the name is upper-cased, characters the instrument will not take are removed, the stem is clipped to eight characters and the extension to three, and a clash is given a ~1 tail. The drop plan lists every conversion before anything is written.*

How Long an Upload Takes

The payload crosses the bus twice, once to write and once to verify. Measured over GPIB, a TDS 784C reads at about 3.9 kbytes/s and a TDS 640A at about 3.2 kbytes/s from a floppy. The drop plan estimates the time before anything is sent.

Creating Folders

Press **New folder** and give it a name. The name is checked against the same 8.3 rule, checked against what is already in the folder, upper-cased, and sent as `FILESYSTEM:MKDIR`.

Deleting Files and Folders



WARNING. *There is no recycle bin on the instrument. A file deleted here is gone, and a folder is removed with everything inside it. Download anything you may want before deleting it.*

Select what is to go and press **Delete**. Which pane has focus decides what is acted on, as in Explorer. Folders and files go separately, and folders one at a time, so what is about to be removed can be shown properly first.

A folder is surveyed first and the confirmation names what is inside it. Every file is checked before any is deleted, so a selection containing something protected removes nothing.

What Cannot Be Deleted

The instrument runs its applications from a Java runtime held on the same disk, and the program refuses to delete any of it. Nothing on a floppy is protected.

Table 2-2: Protected names on the instrument's hard disk

Name	Why
APP	Holds the Java runtime and every shipped application
APP/TDSRTE1 and everything below it	The Java runtime itself
STARTUP.BAT, OSSA.BAT, RTE1.BAT, RTE1ORIG.BAT	The instrument's boot chain. Deleting one stops the runtime starting at power-on
RT.JAR, JAVA68K.O, LIBJIT.O, NIGPIB.O, EXTCP.O, PATCH.O, LOGO.BIN, VERSION.DAT	Part of the Java runtime. Putting one back means reimaging the card

What the Program Sends

A file is deleted with `FILESYSTEM:DELETE` and a folder with `FILESYSTEM:RMDIR`. `DELETE` does nothing to a directory: no error, no event, just silence and the directory still there. `RMDIR` is recursive and silent, with no dry run.



CAUTION. *The program moves the working directory to the volume root before it sends RMDIR. RMDIR issued while the instrument is standing inside the folder being removed is refused with event 257, and that refusal has been seen to leave the filesystem subsystem returning a fixed garbage pattern for every path until the card was reimaged.*

Both are verified by re-listing the folder afterwards rather than by trusting an empty event queue.

Instruments With No Filesystem

Firmware v2.x has no filesystem commands at all: no working directory, no directory listing, no free space. On such an instrument the program says so when it connects and leaves the Files tab empty. The Waveforms tab still works, which on those instruments is what the program is for.

On the A and B series the filesystem is there but `FILESYSTEM:WRITEFILE` is not, so **Upload** is greyed for the session and the reason is given once. Browsing, downloading, creating folders and deleting all work normally.

Screenshot

There is no command on these instruments that says give me the display. What there is, is a hardcopy subsystem meant for a printer, which will send to the bus if it is asked. A screenshot is therefore a print job aimed at this program, decoded here and drawn on the screen.

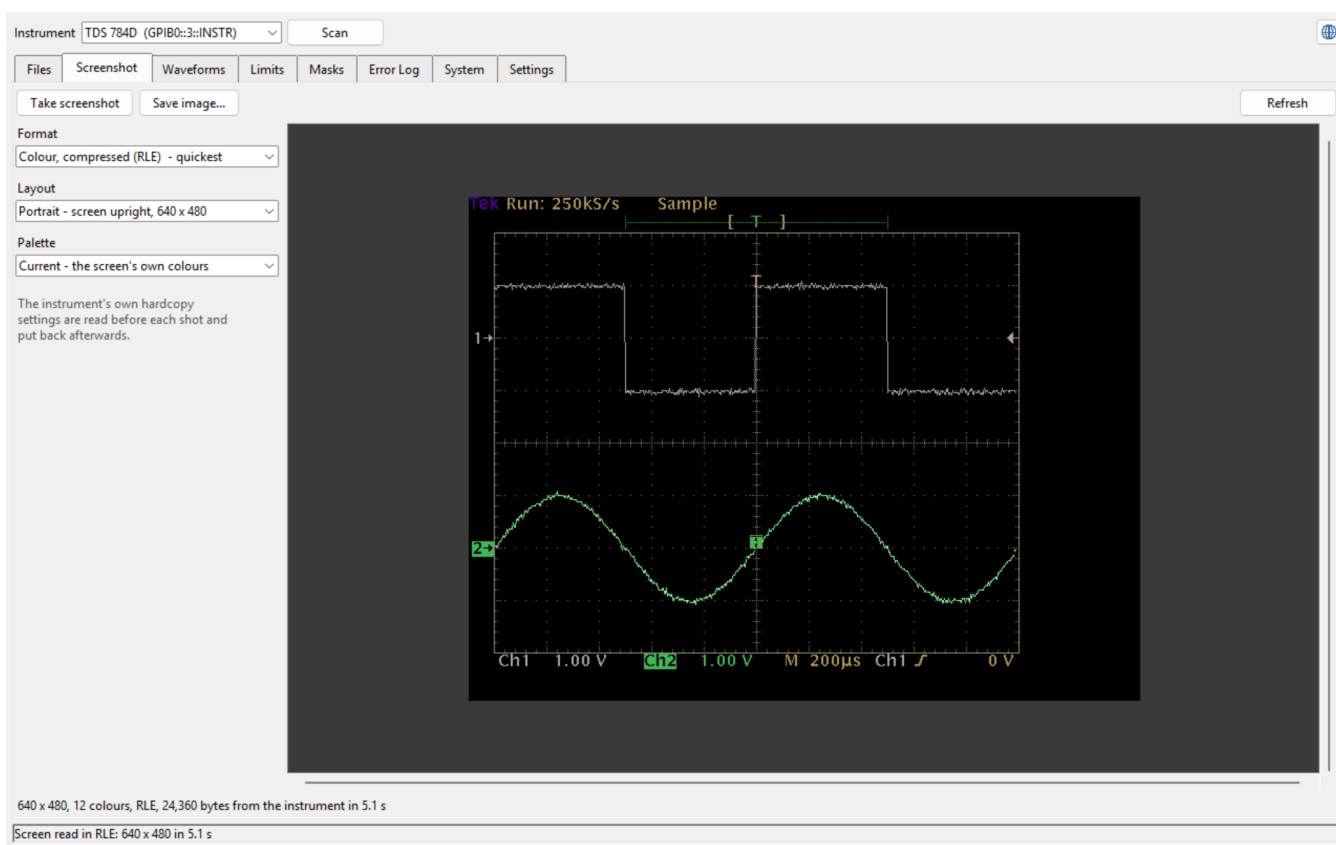


Figure 3–1: The Screenshot tab

Taking a Screenshot

1. *Choose the format:* The **Format** box lists only what this instrument accepted when it was asked. The best available is already chosen.

2. *Choose the layout:* **Portrait - screen upright, 640 x 480** gives the screen the right way up. **Landscape - screen turned 90 degrees** turns it on its side, at 480 x 640.
3. *Capture:* Press **Take screenshot**. The picture appears in the pane and the line under it says what arrived.
4. *Save it:* Press **Save image...** and choose a name.

The line under the picture gives the size in pixels, the number of colours, the format the instrument used, how many bytes arrived and how long they took. The colour count is the number of palette entries the picture actually uses, not the size of the palette.

Formats

The instrument's hardcopy subsystem offers printer languages as well as image formats. Only the formats that are images, and that this program can decode, are listed. Which they are is established by setting each one and reading it back, because a monochrome instrument accepts BMPCOLOR without complaint and then reports BMP.

Table 3-1: Image formats

Shown as	Keyword	Colour
Colour, compressed (RLE) - quickest	RLE	yes
Colour PCX	PCXCOLOR	yes
Colour, uncompressed (BMPCOLOR)	BMPCOLOR	yes
Monochrome (TIFF) - quickest	TIFF	no
Monochrome (BMP)	BMP	no
Monochrome PCX	PCX	no

Almost all of the time a capture takes is the instrument drawing the picture, so the smallest file is not the quickest. Measured over GPIB with a National Instruments GPIB-USB-HS:

Table 3–2: Measured capture size and time

Format	TDS 784D bytes	TDS 784D s	TDS 640A bytes	TDS 640A s
TIFF	38,830	3.28	38,858	3.27
BMP	38,462	4.35	38,462	6.14
PCX	17,763	4.40	14,587	6.43
PCXCOLOR	81,351	5.30	no colour formats	
RLE	85,318	5.35		
BMPCOLOR	308,278	5.62		

RLE is chosen by default where the firmware has it: it is colour, it is a quarter of BMPCOLOR's bytes, and it is a Windows BMP with run-length encoding, so it needs no separate decoder. Where there is no colour at all, TIFF is chosen, being twice as quick as BMP on a 640A for the same pixels.

Layout and Palette

Layout

LAYOUT is the page's orientation, not the image's. A change here turns the picture already in hand rather than waiting for another capture. A 640A found set to LANDSCAPE produced a 480 x 640 image lying on its side; choosing **Portrait** puts it upright.

Palette

What the **Palette** box offers depends on the instrument. A colour instrument has a hardcopy palette of its own and is offered its two settings; a monochrome instrument has none, and is offered this program's own inversion instead.

Table 3–3: Palette choices

Instrument	Choice	What it does
Colour	Current - the screen's own colours	The instrument sends what the screen shows
Colour	Hardcopy - light background, for printing	The instrument sends its printing palette
Monochrome	Normal - dark on light, for printing	The picture as the instrument sends it
Monochrome	Inverted - as the screen shows it	Each palette entry subtracted here

NOTE. On a colour instrument the palette belongs to the instrument, so the choice applies to the next capture and the picture on screen does not change. On a monochrome instrument the inversion is done here and the picture changes at once, which avoids repeating a five-second capture.

Saving the Picture

The file type list offers the instrument's own format first and PNG second. Either way what is written is what the pane is showing, including any rotation and any inversion.

NOTE. *A picture nothing has been done to is saved in the native format as the bytes the instrument sent, exactly. One that has been turned or inverted here is written out again in the same family and at the same depth it arrived in, which is a new file rather than the original. A picture that arrived compressed is written uncompressed.*

The suffix offered for the native format is taken from the bytes that arrived rather than from the format that was asked for, so an instrument that answers TIFF and sends BMP does not produce a mislabelled file. The default file name is stamped with the date and time, in the order year, month, day, so that a folder of them sorts chronologically.

What the Program Sends

The instrument's whole hardcopy configuration is read first, the capture is taken, and the configuration is written back afterwards. All three bench instruments arrived set to print to a floppy and are entitled to be left that way.

```
HARDCOPY?                               (every field, read first)
HARDCOPY:PORT GPIB
HARDCOPY:FORMAT <keyword>
HARDCOPY:LAYOUT <layout>
HARDCOPY:PALETTE <palette>             (only where the field exists)
HARDCOPY START                          (then read until EOI)
```

Two waits are not politeness. A hardcopy started too soon after the previous one is ignored, with no data and no error, so two seconds are allowed to pass before START. The VISA timeout is raised to ninety seconds for the read itself.



CAUTION. *If a capture fails, the program issues a device clear and waits a second before reporting it. A half-sent hardcopy left in the instrument is what makes the next program's *IDN? time out.*

The picture is decoded by sniffing the first bytes rather than by trusting the format keyword. BMP at 1, 4 and 8 bits, uncompressed or run-length encoded, PCX at 1 or 8 bits, and the one baseline uncompressed form of TIFF a TDS writes are all handled. Anything else is refused by name.

NOTE. *The instruments write a file-size field in the BMP header that is not the file's size: 77,070 in a 308,278-byte BMPCOLOR. Nothing in the decoder reads it.*

Waveforms

The Waveforms tab captures what the instrument is acquiring, plots it, saves it, and writes it back. It works on every instrument in the range, including the v2.x models that have no filesystem at all, because the waveform transfer commands are present in every firmware image examined.

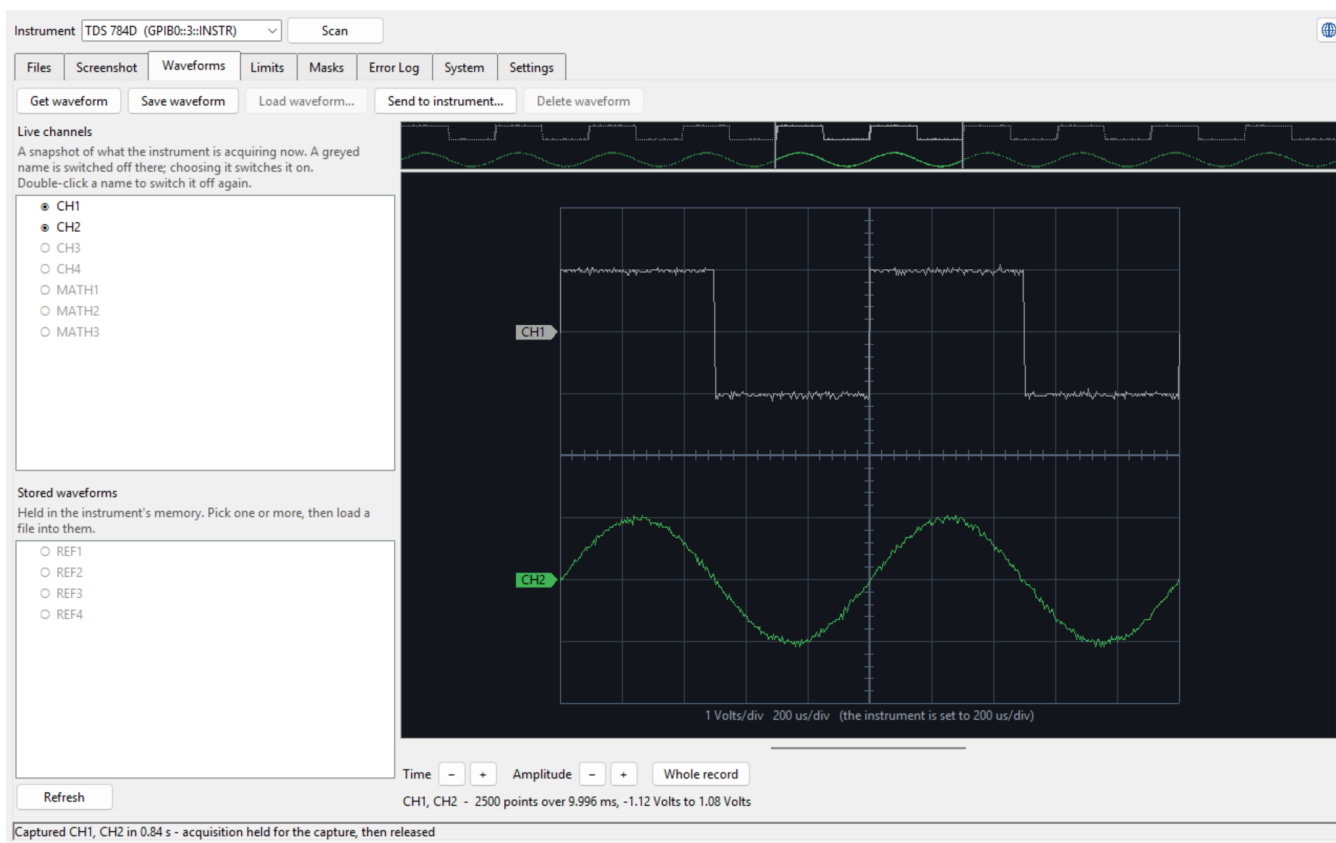


Figure 4–1: The Waveforms tab, showing two captured channels

The Source Lists

The instrument's sources are listed in two panes because they behave differently. A channel is acquired: every read is a fresh snapshot. A reference is stored: every read gives the same samples.

Live channels	Every channel and maths waveform the instrument reports. A name in grey is switched off at the instrument.
Stored waveforms	The reference memories the instrument reports, normally REF1 to REF4.

Choosing a source is a click on the glyph at the left of its row, which fills in. Double-clicking a name switches that source on or off at the instrument.

NOTE. *A channel that is not displayed cannot be read. Asked for one, the instrument answers event 2241, "waveform requested is invalid", and follows it with a query interrupt. Choosing a greyed name therefore switches it on at the instrument first. A stored reference is the exception and reads whether it is displayed or not.*

Capturing

Press **Get waveform**. Every live source that is chosen and displayed is read from one acquisition, and any chosen references are read after it.

The acquisition is stopped for the read and released afterwards. This is not tidiness. Reading CH1 and then CH2 while the instrument is running gets two different captures: measured on a 784D triggering thirteen times a second, the two reads were 600 ms apart and disagreed about XZERO. Stopped, the two channels report the same XZERO and the same source read twice comes back byte for byte the same.

Three things are deliberately not done:

- An instrument that was already stopped is left stopped.
- An instrument set to single sequence, or set to stop on a limit test, is not touched: stopping it would re-arm the trigger, and with STOPAFTER LIMIT there is nothing for an operation-complete query to complete, so the bus would wait out its timeout.
- References are read after the acquisition is released, since they are not moving.

The status line says which of the three happened, for example "Captured CH1, CH2 in 0.85 s - acquisition held for the capture, then released".



WARNING. *Waveform capture is refused while the instrument is in DPO, and the program does not switch DPO off. In DPO the instrument builds a pixel database instead of filling a waveform record, and every point of every channel comes back as 128. Switching DPO off would discard the accumulation on the screen, which is the very thing being looked at. Take a screenshot instead, or disable DPO mode yourself.*

Reading the Plot

Traces are drawn at the instrument's own scale: a sample is placed at its own count divided by the counts in a division, which is exactly where the instrument puts it on its own screen. A small signal on a large volts-per-division setting therefore draws small, the way it looks on the scope, rather than being stretched to fill the graticule.

Record strip

The band above the graticule shows the whole record with the visible window outlined, the way the instrument's own record view does.

Channel marker

The arrow outside the graticule sits at that trace's zero volts and carries its name. Markers that would collide step aside, and each points towards the graticule.

Caption

Under the graticule, the volts and seconds a division of the view. Where the view is longer than the instrument's ten divisions, the instrument's own setting is given as well.

Table 4-1: Mouse and keyboard on the plot

Gesture	Effect
Drag	Pan. Whatever is under the pointer stays under it
Drag a marker	Move that trace only, up and down, like the instrument's position knob. It snaps onto the centre line within half a minor division
Shift and drag	Rubber band. Zoom to the rectangle on release
Wheel	Zoom in time about the pointer
Shift and wheel	Zoom in amplitude about the pointer
Click the record strip	Centre the view there
Right-click a marker	Hide that trace, or show only that trace

Zoom and Pan

The minus and plus buttons beside **Time** and **Amplitude** step through a 1-2-5 ladder, which is the ladder the instrument's own knobs use. **Whole record** puts everything back.

Saving a Waveform

Save waveform writes the trace in hand. The format follows the extension:

Table 4–2: What each format keeps

Extension	What is written
.csv	A header line naming the units, then scaled x,y pairs. Read back it is re-quantised, and is correct to within one step of the instrument's eight-bit range
.isf	The instrument's own transfer format: the preamble fields and the curve exactly as CURVE? returned them. Read back it is byte for byte the same
.tdw	The instrument's own layout, under this program's own extension. Eight-bit captures are widened to sixteen bits. See the note below
.png	The graticule as it is on screen now, at the current zoom and with the record strip above it, at the resolution set on the Settings tab and in the current colours

NOTE. A .WFM is a file an instrument wrote. What this program writes is the same layout but is not one of those, so it is given its own extension and the two are never confused. Both are read. To put a real .WFM on the instrument's own disk, use **Send to instrument...**

NOTE. The .WFM format is not documented by Tektronix. What is written here was worked out by comparing the instrument's own files against Tektronix's CNVRTWFM converter, one field at a time, and every field the program reads and writes agrees with that converter on every file measured. Twelve bytes of the 132-byte header are still not understood; they are carried over from a file the instrument wrote rather than invented.

Loading a Waveform

A file is loaded into a reference, and nothing is written to the instrument until **Send to instrument...** is pressed. That way a set of references can be filled in one go, and a file can be looked at without committing it to anything.

1. *Choose the reference:* Click the glyph beside REF1, or beside several references at once.
2. *Open the file:* Press **Load waveform...** and choose an ISF, CSV, TDW or WFM file. The row then reads, for example, "REF1 - trace.isf, not sent yet".
3. *Send it:* Press **Send to instrument...** and confirm.

The file's kind is established from its contents rather than from its extension. A waveform file from a TDS5000, TDS6000, TDS7000 or a DPO or MSO is a different and much later format and is refused by name rather than misread.

Sending a Waveform to the Instrument

Send to instrument... writes the trace in hand into a reference memory, or onto the instrument's disk where the firmware has WRITEFILE. The destination is asked for:

- **Reference memory REF*n*.** The waveform is written into that reference and read back to verify it.
- **A .WFM on the instrument's disk, which the instrument can recall.**

Only the instrument's own format goes onto its disk. An .ISF written there is a file the instrument itself can do nothing with, and fetching one back to a computer over GPIB is a round trip when **Save waveform** writes the same file straight to the computer's own disk.



WARNING. *Whatever the destination reference holds now is replaced. There is no undo, and the instrument gives no warning of its own.*

A reference that does not exist yet cannot be written to. Every field describing the waveform is answered with event 2241, and the curve is met with event 532, "curve data too long, curve truncated"; nothing warns you and the reference stays empty. The program therefore brings the reference into being first with **SAVE :WAVEFORM** from a live channel, which was measured byte-identical on a 784D, a 784C and a 640A.

Every send is read back and compared. A mismatch is reported as "*REF1* was written but reading it back gave something else - the instrument did not keep what was sent."

Deleting a Reference

Delete waveform empties the chosen reference with **DELETE :WAVEFORM**. Only a stored reference can be deleted; a channel or maths trace is computed live and has nothing stored to remove, which is why the button is greyed unless a reference is chosen.

What the Program Sends

Reading one source:

```
DATA:SOURCE <source>
DATA:ENCDG RIBINARY
DATA:WIDTH <1 or 2>
DATA:START 1
DATA:STOP 1000000
HEADER ON
WFMPRE:<source>?           (the whole preamble at once)
HEADER OFF
DATA:START 1
DATA:STOP <NR_PT>
CURVE?
```

Two orderings in that sequence are not arbitrary:

- **DATA:SOURCE** must be written before any **WFMPRE** query, or the instrument answers event 2241 even for a reference that demonstrably holds a waveform.
- **NR_PT** does not report the record length. It reports how many points would be sent, which is **DATA:START** to **DATA:STOP**. A 784D was measured reporting **NR_PT** 20 for a 5000-point record, because something earlier had set **DATA:STOP** 20.

The preamble is asked for with the headers on, so that every field names itself. The headers-off reply is not in the same field order on every instrument, and mapping one instrument's order onto another puts a unit string where a sample count belongs. Named, the order does not matter. Measured against asking the eleven fields one at a time, 419 ms becomes 81 on a 784D, 457 becomes 76 on a 784C, and on a 640A 3510 becomes 196, because that instrument has no XZERO and does not answer the query for it at all.

NOTE. *Where the bulk reply comes back in some shape the program does not recognise, it falls back to asking the eleven fields one at a time. One instrument in thirty-four answering unexpectedly should cost a little speed, not the capture.*

Limits

A limit test judges every acquisition against an upper and a lower limit and stops the instrument the moment the signal leaves them. The instrument does the judging; this tab builds the template, starts the test and reports the verdict.

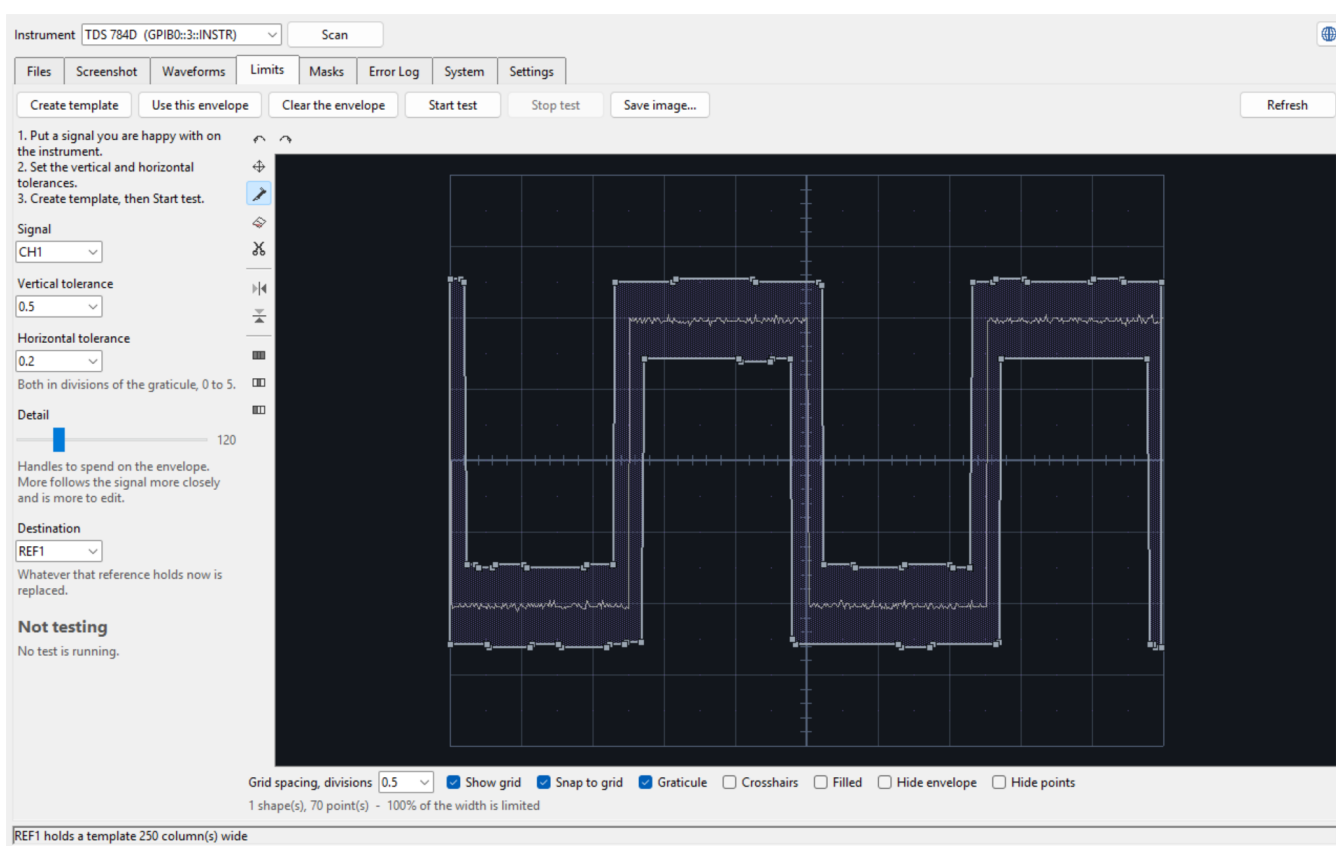


Figure 5–1: The Limits tab, showing a template learnt from a square wave

Running a Limit Test

1. *Put up the signal:* Set the instrument up so that it is showing a signal you are happy to call good.
2. *Choose the signal and the destination:* Set **Signal** to the channel to be judged and **Destination** to the reference channel the template is to be written into.

- 3. *Set the tolerances:* Enter how far the signal may wander, vertically and horizontally, in divisions of the graticule.
- 4. *Make the template:* Press **Create template**. The instrument builds the template and the tab draws it.
- 5. *Start the test:* Press **Start test**. The verdict reads PASS while the instrument is running.
- 6. *Stop the test:* Press **Stop test** to switch the test off and let the instrument run freely again.



WARNING. *Whatever the destination reference holds now is replaced, both by **Create template** and by **Use this envelope**. Save anything you need from that reference first.*

Tolerances

Both tolerances are in divisions of the graticule, from 0 to 5, which is the range the instrument's own command takes. Half a division is 0.5. A tolerance of zero is taken without complaint and gives a template the width of the signal itself.

NOTE. *The tolerance is applied by the instrument when it builds the template, not by this program. Changing a tolerance after a template has been built has no effect until it is built again.*

The Verdict

The instrument reports a failure by stopping. That is the whole of the verdict, and it is what this tab watches for: while the acquisition is running the test is passing, and the moment it stops the signal has left the template. The tab asks every two seconds, and nothing in the asking writes to the instrument.

Table 5–1: What the verdict means

Verdict	State
Not testing	No test is running
PASS	The signal is inside the template and the instrument is still acquiring
FAIL	The instrument stopped, so the signal left the template

Drawing the Envelope

The envelope a signal is allowed into can be drawn by hand instead of built by the instrument, or a built one can be adjusted. The editor is the same one the Masks tab uses and is described in Section 6; the difference is what the drawing means.



CAUTION. On this tab the envelope is where the signal may go. On the Masks tab the same shape means the opposite: where the signal may not go. Sending one for the other passes everything it was drawn to catch.

Press **Use this envelope** to write what is drawn into the destination reference. **Clear the envelope** takes the drawing off the canvas and leaves what the instrument is holding alone.

A limit test allows the signal one band. A drawing that leaves two bands in a column, with clear air between them, cannot be expressed, and the program says which columns and offers to close the gap up.

NOTE. A drawing on this tab may hold up to 64 shapes of up to 200 points each. Those are not instrument limits: the drawing becomes one envelope, as wide as the record is long, whatever it is made of. They are there to stop a runaway, and because of how many handles a person can work with.

What a New Template Leaves on the Canvas

A template holds two values for every column of the record, so a 500-point record gives 250 columns and five hundred handles, which is far more than anyone can edit. The program thins it to as many handles as the **Detail** slider asks for - 120 to begin with, half to each edge, and anything from 20 to 500 - and leaves the instrument's own band drawn underneath so the cost of the thinning can be seen. Dragging the slider re-thins from that band rather than from the drawing, so the choice can be seen rather than guessed at.

The thinning only ever loosens. Thinned plainly, the upper edge cuts every corner inwards and takes away exactly the room the tolerance was there to give.

What the Program Sends

```
ACQUIRE:STOPAFTER RUNSTOP
LIMIT:STATE OFF
DISPLAY:MODE NORMAL
ACQUIRE:STATE RUN
*OPC?
LIMIT:TEMPLATE:SOURCE <source>
LIMIT:TEMPLATE:DESTINATION <dest>
LIMIT:TEMPLATE:TOLERANCE:VERTICAL <divisions>
LIMIT:TEMPLATE:TOLERANCE:HORIZONTAL <divisions>
LIMIT:TEMPLATE STORE
```

Create Template

The tolerances are set before STORE, because STORE is what reads them. The display is taken out of DPO first: a limit test under InstaVu reads back as on and tests nothing, answering event 550, "InstaVu active", and a template built there would be built from a record that does not exist.

What comes back really is an envelope. Measured on a 784D, the reference reported "Ref3, DC coupling, 500.0mVolts/div, 200.0ns/div, 500 points, Envelope mode", and a test against it held while the signal was unchanged and stopped within two seconds of the amplitude being doubled.

```

ACQUIRE:STOPAFTER RUNSTOP
DISPLAY:MODE NORMAL
ACQUIRE:STATE RUN
*OPC?
LIMIT:COMPARE:<source> <dest>
LIMIT:STATE ON
*OPC?
LIMIT:STATE?                (read back to confirm)
ACQUIRE:STOPAFTER LIMIT
ACQUIRE:STATE RUN          (and no *OPC? after it)

```



CAUTION. Nothing may ask *OPC? after ACQUIRE:STOPAFTER LIMIT. With the acquisition told to stop only when the test trips, there is nothing for an operation-complete query to complete, and the bus waits out its timeout.

Start Test

LIMIT:STATE is read back rather than assumed. An instrument that will not take the test on says so by leaving the state off, and its own explanation is drained from the event queue and shown.

Reading the Template Back

Refresh reads the template out of the reference and the live signal off the channel, and draws both. Two orderings matter, and both were measured:

- The preamble must be read after DATA:WIDTH 2, because WFMPRE answers for the width currently set. Read first, YMULT comes back as the eight-bit 20 mV a count rather than the sixteen-bit 78.125 uV, and every volt in the band is 256 times too big, which draws a band covering the whole graticule. A band that covers everything passes everything.
- DATA:STOP counts values, not points. Asked for twice NR_PT the instrument answers event 531, "Data stop > record length, Curve truncated", and sends NR_PT of them anyway, giving half the columns spread over the whole record. Measured against a 1 kHz square at 200 us/div, the edges landed 62.3 columns apart out of 250, which is 498 us where 500 is right.

Sending a Drawn Envelope

A drawn envelope becomes an upper and a lower value for each column, loaded into the reference as an envelope. The scaling fields are copied from the channel's own preamble, because a template written against a 1 V/div channel came back as 500 mV/div otherwise, and because a channel sitting at YOFF 12800 needs the band built about 12800 or it lands two divisions out and fails a good signal.

NOTE. Reading an envelope back off an instrument has not been measured on one. The program reports a mismatch on the status line rather than raising it as an error, because the read-back itself is not trusted enough to call the instrument wrong.

Masks

A mask is a keep-out area drawn in percent of the graticule. The instrument holds eight of them, draws them on its screen, and counts how many acquisitions touch them. This tab keeps a library of masks on the computer, edits them, sends them to the instrument, and sets the instrument up to judge a signal against one.

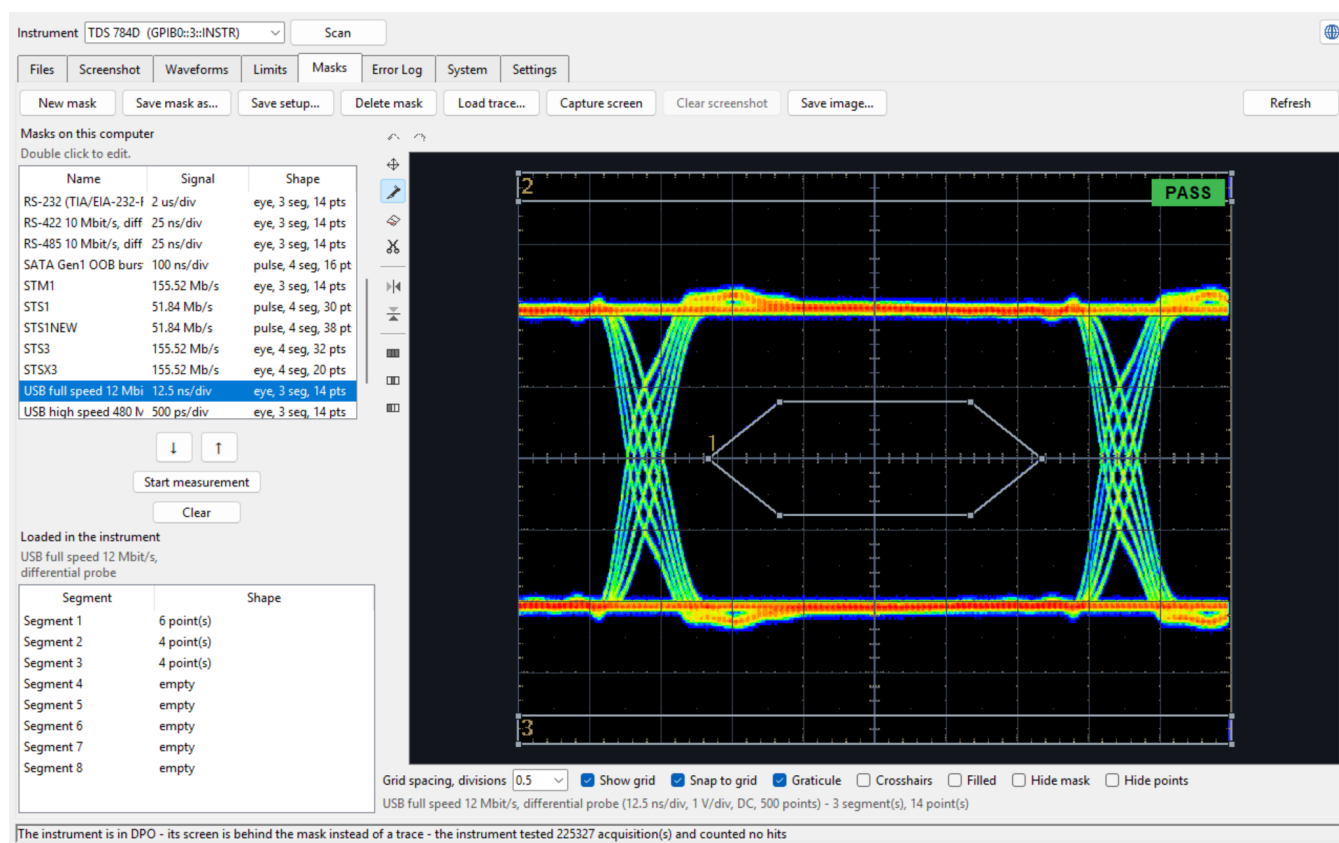


Figure 6–1: The Masks tab, with a USB full speed mask open

NOTE. The mask subsystem is Option 2C. Without it the instrument answers the mask queries and draws nothing, so **Send to instrument** and **Start measurement** are greyed for the session. Drawing, editing, saving and loading are not: they need no instrument at all. To judge a signal on an instrument with no 2C, use the **Limits** tab, which reaches a verdict by a different means. See Section 5.

The Library

The upper list on the left is the library of masks on this computer, kept in a `masks` folder beside the program. The masks supplied with the program are copied there the first time it runs. Double-click a row to open that mask in the editor.

Table 6-1: What the library lists

Column	What it shows
Name	The mask's own name, from inside the file
Signal	The standard's bit rate where the mask is one Tektronix shipped, otherwise the timebase from the setup beside it
Shape	Whether it is a pulse mask or an eye mask, and how many segments and points it holds

The lower list shows the eight segments the instrument itself is holding, and the line above it names what was last sent from here. Between the two lists are the controls that move a mask either way.

Down arrow Send the open mask to the instrument's eight segments.

Up arrow Read the instrument's eight segments back into the editor.

Start measurement Set the instrument up to judge this mask and start counting.

Clear Empty the instrument's eight mask segments.

NOTE. *There is no mask file on the instrument. There is no `SAVE:MASK` and no `RECALL:MASK`; the eight segments are all the instrument has, and the library lives on the computer.*

Drawing

The Tools

Coordinates are in percent of the graticule, X across and Y up from the bottom, so the whole graticule is 0 to 100 in both. One division is 10 percent of the width and 12.5 percent of the height, which is why the grid steps differ in the two axes. The tools are a column of buttons to the left of the canvas, drawn as icons and named only by the note that appears when the pointer rests on one. They are listed here in the order they appear, top to bottom. The first four are modes, exactly one of which is in hand; the rest act at once on what is selected.

Table 6–2: The tool palette

Icon	Named	What it does
Four-way arrow	Select and move points and shapes	Drag a point to move it, or an edge to move the whole shape. Arrow keys nudge what is selected
Pen	Draw points	Click to place points. Right-click, Enter or Escape finishes the shape
Eraser	Delete points and shapes	Click a point to delete it, or an edge to delete the whole shape
Scissors	Split a shape between two of its points	Click one point of a shape, then another, to cut it in two between them
Two wedges	Mirror the selection horizontally, or vertically	Flip what is selected about its own axis
Two squares	Union, Intersect, Subtract	Combine two chosen shapes. The order matters: subtract takes the second out of the first

Choose a shape with the move tool, then hold Shift and click a second one to choose both. The status line under the canvas always says what the tool in hand does.

Table 6–3: Keys in the editor

Key	Action
Escape, Enter	Finish the shape being drawn, or abandon a cut
Delete, Backspace	Delete the selected points. Two neighbouring points of one shape collapse to their midpoint
Arrow keys	Nudge the selection by a fifth of the grid step, or by the distance set on the Settings tab
Ctrl+C, Ctrl+X, Ctrl+V	Copy, cut and paste whole shapes
Ctrl+Z, Ctrl+Y	Undo and redo, twenty deep

NOTE. Up is up. Percent counts from the bottom of the graticule here; the instrument numbers its own mask points from the top, and the flip between the two happens once, on the way out.

Typing Coordinates

Double-click a point to open the Coordinates dialog for its shape. The dialog lists every point with its X and Y to two decimal places, and carries the shape's midpoint at the top. Typing into either moves the shape or the point as the characters are typed, and the whole edit is one undo step. While the dialog is open the points are numbered on the canvas.

Right-clicking opens the same dialog, and is the way to reach it for a whole shape rather than a point. Right-click a point for **Delete** and **Coordinates** for that point; right-click a side or the inside of a shape for the same two, acting on the shape. A shape's coordinates are its bounding box's middle, so typing 50, 50 there puts the shape in the middle of the graticule.

***NOTE.** While a shape is being drawn, neither of those happens: a double-click or a right-click finishes the shape instead. The second click of a double-click lands on the point the first one laid down, which is not a request to edit it.*

Options Under the Canvas

Grid spacing, divisions	The grid step, in divisions.
Show grid	Draw a dot at every grid crossing.
Snap to grid	Whether clicks, drags and nudges snap. Independent of Show grid.
Graticule	Draw the instrument's 10 by 8 graticule.
Crosshairs	Lines that follow the pointer, with a readout.
Filled	Draw the shapes solid rather than outlined.
Hide mask	Hide the shapes so the trace behind can be seen. Switched off again whenever a tool is picked up.

What the Instrument Will Take

The instrument's mask subsystem has hard limits, and a mask that breaks one cannot be saved or sent until it is corrected:

Table 6–4: Mask limits

Limit	Value
Segments	8. The instrument answers MASK1 to MASK8; MASK9 and beyond give nothing
Points on segment	50
Minimum points on segment	2. A single point comes back as 0,0
Coordinates	0 to 100 percent in both axes

Separately, the instrument stores the points exactly and joins them up its own way. Three arrangements come back drawn differently from the way they went in, and the program warns about each before sending:

- Three points in a line one above the other: the instrument drops one of them.
- A shape a vertical line can cross more than twice.
- A shape a horizontal line can cross more than twice.

Concavity on its own is not a problem. Cutting the shape into simpler pieces avoids all three, and a segment in this state is drawn dashed on the canvas.

Sending a Mask

Press the down arrow. The dialog that opens states where the mask is going and what will be replaced.



WARNING. *Sending replaces whatever mask the instrument is holding, all eight segments of it. What is on the computer is untouched.*

Where a setup file sits beside the mask, the program offers to send that first. The setup names the channel, the volts a division and the timebase the mask was drawn against, and without it the mask is judged against whatever the instrument happened to be set to.

A setup is read back after it is sent and compared with what was asked for. Where the instrument quietly gave something else, the program says which commands and offers to stretch the mask to mean what it meant before, because a mask is held in percent of the graticule and can be stretched to match. Where the scale asked for was out of the channel's reach, it says which probe would reach it.

Judging a Signal

Capture screen puts what the instrument is showing behind the mask: a trace where it can be read, or the instrument's own screen where the instrument is in DPO and there is no waveform record to read. **Load trace...** puts a captured or saved waveform there instead.

A verdict badge is drawn at the top right of the graticule. Where the instrument's own counter has a tally, that is the verdict; failing that, the drawn samples are tested against the shapes here. An empty mask is not a pass, and nothing is shown for one.

Setting Up an Eye

An eye mask is judged against a folded eye diagram, and the instrument has to be set up for one: the sweep slid so the middle of a bit lands on the mask, DPO on where it is fitted, and the hit counter started. **Start measurement** does all of it.

Two things cannot be discovered and are asked for first.

Start measurement

The instrument will be set up for an eye: the sweep slid so the middle of a bit lands on the mask, DPO on where it is fitted, and the hit counter started.

This signal is probed

☒ with one differential probe, into a single channel

☐ with two probes on CH1 and CH2, differenced in MATH1

MATH1 is defined as CH1-CH2 and judged here rather than by the instrument: its mask counter takes a channel and refuses a math waveform, and DPO switches math off, so that route runs on persistence. The instrument also scales a math waveform itself - there is no command for it - so on its own graticule the difference will not line up with the mask. The picture here does.

and triggered from

☒ a clock at the bit rate, on CH2

☐ the data itself

The named input is switched on just long enough to check there is a clock at the bit rate on it, and the sweep is then slid so the middle of a bit lands on the mask. Triggering on the data gives a pseudo-eye: the instrument's own time base jitter is added to the signal's, and the sweep is delayed past the trigger so that both rails reach the middle of the screen.

Cancel Start

Figure 6–2: The Start measurement dialog for an eye mask

How it is probed	Whether the signal arrives on one differential probe or as two probes to be differenced in MATH1. Asked only where the mask's setup says the mask is drawn for a differential pair.
What it is triggered from	A clock at the bit rate on a named input, a clock to be looked for, or the data itself.

Neither can be found out by asking the instrument: it cannot see what is on the end of a probe, and an input with nothing plugged in reads the same as one whose clock is switched off.

With a Clock	The named input is switched on just long enough to check that there is a clock at the bit rate on it, and switched off again. A square wave at the bit rate crosses its own middle every half a bit, evenly; data crosses at whole multiples of a bit and unevenly, and the gaps tell the two apart. Naming the input skips the hunt, not the check. An input with nothing on it is reported rather than triggered on, because the instrument would sit waiting for an edge that is never coming. The trigger level is measured rather than set to fifty percent, because there is no set-to-50% on this firmware: <code>TRIGGER:MAIN:SETLEVEL</code> answers event 113, "Undefined header". The clock's own trace is then switched off. Measured on a 784D with a 2 Vpp clock left displayed, its trace across the keep-out scored 2,143,642 hits over 58,455 acquisitions on a signal that passes cleanly the moment it is switched off.
Without a Clock	Triggering on the data can only ever show half an eye: the bit after a rising edge is always high. The instrument's own answer is the delayed time base, and the program uses it: trigger on a data edge and look sixteen bits past it, by which time the data no longer remembers which edge it was and ones and zeroes both land in the middle. The delay is then tuned against the mask itself.
Checking the Sweep	An eye wants about one bit on the graticule. The program measures how many the sweep is holding and says so if it is far off, because a screen holding twelve bits, or a twelfth of one, is not going to make an eye. Measured on a 784D, a USB full speed mask against 1 us/div gave 116,921,796 hits over 345,839 acquisitions and hid the clock as well.

NOTE. On the two-probe route the instrument's own counter is not used. `MASK:SOURCE` will not take `MATH1`, so the counter would be tallying one leg of the pair against a mask drawn for the difference. That route runs on persistence and is judged here instead.

Mask Files

Two formats are read and written, both with the extension `.MSK`, and which one a file is is established from its contents. Saving keeps the format the file came in as.

This program's format	Plain text: a TDSMASK header, the name, the source, and one line per segment holding its points. Keeps a shape exactly, including a concave one.
Tektronix i-Pattern	The binary format TTiP used, and the only Tektronix mask format for this generation. Its count field allows ten masks of 50 points, and the 2023-byte file every shipped mask uses has room for nine.



CAUTION. An i-Pattern file keeps a mask's points without their order, so a concave shape does not come back the same way round. The program orders the points by angle about their middle, which recovers a convex shape exactly and a concave one only by luck. It warns before saving a concave shape in that format.

A setup file may sit beside a mask, under the same name with the extension .SET, the way Tektronix shipped one with every mask. **Save setup...** reads the instrument's current settings, shows them, and writes one.

The dialog is a starting point rather than the answer. A mask is often drawn for a signal nobody has on the bench yet, and the setup beside it has to say what the mask means and not what happened to be on the screen, so every field can be changed before it is written. Anything left empty is not written at all.

Unit interval

The bit period the mask is drawn for, in seconds. This is what makes a mask an eye mask, and the eye setup cannot be done without it. No instrument setting records it, so it can only be typed here.

The signal is differential

Whether the mask is drawn for a signal across a pair. It decides what Start measurement asks for and how it sets the instrument up.

NOTE. Those two are written as REM lines, which the instrument ignores and this program reads. A setup carrying them is still a setup the instrument will load.

Error Log

The instrument keeps its own error history in non-volatile memory. This tab reads it, saves it, and clears it. It is not the event queue this program watches while it works, and it is not a log of anything this program did.

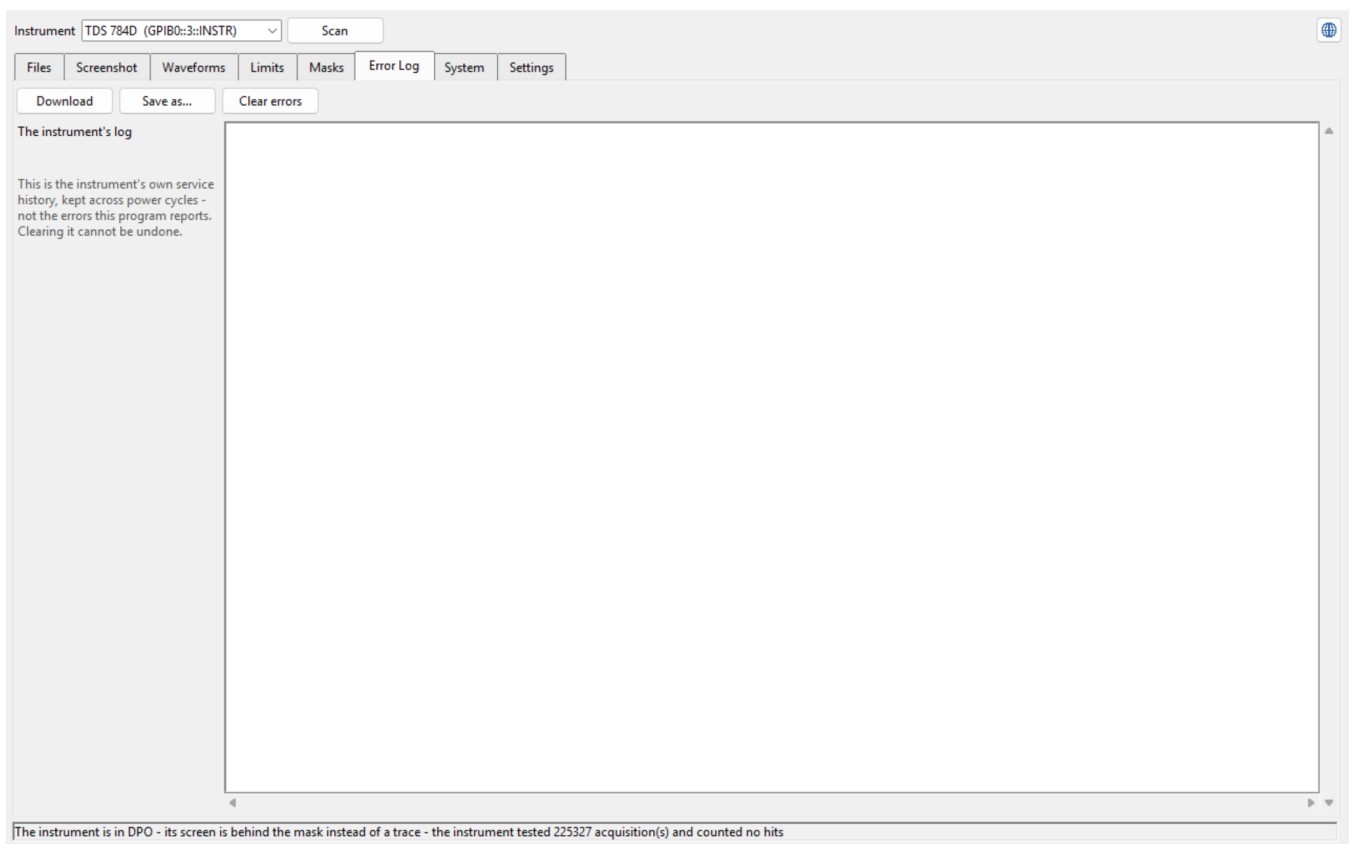


Figure 7-1: The Error Log tab

Reading the Log

Press **Download**. The whole log is read, oldest entry first, and shown in the pane numbered from 1. The text is the instrument's own wording, unaltered: nothing here classifies or filters it.

The pane opens with the instrument's identity and the time the log was read, and closes with "End of errors". A log with nothing in it reads "No errors".

The pane is read-only but can be selected and copied. Arrow keys, Home, End, Page Up, Page Down and Ctrl+C all work.

NOTE. Reading is not instant. There is no command that returns the whole log: it is walked entry by entry, and 31 entries off a TDS 640A took 3.24 seconds. The walk stops after 2000 entries, which is a guard against firmware that never says it has finished rather than a limit anybody should reach.

Saving the Log

Press **Save as....** The file is plain text and is named after the instrument and stamped with the date and time, for example TDS 784D errorlog-20260821-210345.txt.

Clearing the Log



WARNING. Clearing the error log erases the instrument's error history. It is kept across power cycles and cannot be recovered. Download and save it first if there is any chance it will be wanted.

Press **Clear errors** and confirm. The program clears the log, reads it back, and says so. If the instrument still reports entries afterwards it says that instead, with whatever the instrument gave as a reason.

Instruments With No Error Log

Where the firmware has no ERRLOG subsystem the pane reads "This firmware has no error log." and **Download** and **Clear errors** are greyed for the session.

What the Program Sends

There is no command that returns the whole log. It is walked:

ERRLOG:FIRST?	the oldest entry, and positions the cursor
ERRLOG:NEXT?	repeatedly, until an empty reply

Every entry arrives wrapped in double quotes, which are SCPI delimiters and are stripped. An empty log answers with an empty quoted string, not with silence; silence means the firmware has no ERRLOG at all.

Clearing is ERRLOG CLEAR, a value given to the header rather than a subcommand. ERRLOG: CLEAR does not exist and answers event 113, "Undefined header". The event queue is drained before and after.

System

The System tab holds the housekeeping the instrument keeps in non-volatile memory: its identity, the front-panel lock, its clock, the hardcopy and RS-232 ports, signal path compensation, the extended diagnostics, a secure erase, a factory recall, and a factory option unlocker.

The screenshot shows the 'System' tab of the TDS Toolkit software. At the top, there's a header bar with a dark grey background and a white diagonal line. Below it, the 'Instrument' dropdown is set to 'TDS 784D (GPIB0:3::INSTR)' with a 'Scan' button next to it. A tab bar contains 'Files', 'Screenshot', 'Waveforms', 'Limits', 'Masks', 'Error Log', 'System' (selected), and 'Settings'. The main content area is divided into several sections: 'Instrument ID' showing 'TEKTRONIX,TDS 784D,0,CF:91.1CT FV:v7.4e' with 'Read', 'Lock front panel', and 'Unlock front panel' buttons; 'Date and time' with date and time pickers, a checkbox for 'Show the clock on the instrument's screen', and 'Apply' and 'Synchronise to this PC' buttons; 'Hardcopy and ports' with dropdowns for 'Port' (FILE), 'Format' (TIFF), and 'Layout' (PORTRAIT), followed by RS-232 settings (Baud: 9600, Parity: NONE, Stop bits: 1) and an 'Apply' button; 'Calibration and self test' with a description of Signal Path Compensation (SPC), a 'Signal Path Compensation' button, a 'Self test' dropdown set to 'ALL', and a 'Run' button; 'Memory' with a description of Secure Erase, a warning 'THIS ACTION CANNOT BE UNDONE!', and 'Secure erase...' and 'Recall factory setup' buttons; and 'Factory options' with a description and an 'Options...' button. A footer bar at the bottom contains the text 'Read the instrument's settings'.

Figure 8–1: The System tab

NOTE. The tab reads the instrument once, the first time it is brought to the front. A dozen queries at every start-up takes time, and most sessions never open this tab. Press **Read** to read it again, and after changing instruments.

Identity

The first box shows the reply to ***IDN?** and, under it, the reply to **ID?**, which is where the instrument lists the options it believes it has.

Read Read every setting on this tab from the instrument.

Lock front panel Send LOCK ALL. The instrument's own controls stop responding until it is unlocked.

Unlock front panel Send UNLOCK ALL.

Date and Time

The instrument's clock is set in separate boxes rather than as one string, because the instrument takes yyyy-mm-dd and nothing else. A single figure typed for the month is padded to two before it is sent.

1. *Set the clock from this computer:* Press **Synchronise to this PC**. The date and time are taken at the moment of the press.
2. *Or set it by hand:* Type into the boxes and press **Apply**.

Show the clock on the instrument's screen is sent with either, and on its own by **Apply**. A row of empty date or time boxes is not sent: an instrument told 0000-00-00 would take it.

Hardcopy and Ports

The hardcopy port, format and layout are the instrument's own settings, the ones the Screenshot tab borrows and puts back. Setting them here sets them for the instrument's own front-panel hardcopy button.

Table 8–1: Hardcopy and RS-232 settings

Setting	Choices
Port	GPIB, RS232, CENTRONICS, FILE
Format	BMP, BMPCOLOR, TIFF, PCX, PCXCOLOR, EPSIMAGE, INTERLEAF, THINKJET, DESKJET, LASERJET, EPSON
Layout	LANDSCAPE, PORTRAIT
Baud	300, 600, 1200, 2400, 4800, 9600, 19200
Parity	NONE, EVEN, ODD
Stop bits	1, 2

NOTE. The RS-232 settings are Option 13 only. Where the option is not fitted they read back blank, and a blank box is not sent. An instrument that refuses a line says so, and the refusals are collected and shown together rather than stopping at the first.

Calibration and Self Test

Signal Path Compensation

Signal path compensation corrects the instrument's own signal path for the temperature it is at now. Press **Signal Path Compensation** and confirm.



CAUTION. Allow the instrument its twenty-minute warm-up and disconnect every input signal before compensating. A signal present during compensation is compensated into the correction.

Self Test

The command is *CAL?, and the program allows it ten minutes. A reply of 0 is a pass; anything else means the instrument could not compensate itself, and the self test below will say more about why. Nothing else can use the bus while it runs. Choose an area and press **Run**. The areas are ALL, ACQUISITION, CPU, DISPLAY and FPANEL. The result and the instrument's own log of it appear in the pane below.



CAUTION. The extended diagnostics warm-boot the instrument. Whatever is on its screen is lost, and it does not answer the bus for half a minute or more while it restarts. The program waits up to four minutes for it to come back.

NOTE. A line in the result pane reading "pass -- Cal Initialization (see error log)" is a pass. The words in brackets are part of the test's name, not a report of a failure.

Memory



WARNING. A secure erase writes zeroes over every waveform and setup memory and returns all saved setups to factory settings. Calibration data is left untouched. This cannot be undone.

Press **Secure erase...** and confirm. The command is TEKSECURE and the program allows it five minutes. The instrument reports the result as an event: 2285 is a pass, 2286 is a failure to verify the memory afterwards.

Recall factory setup sends FACTORY, which returns the instrument to its default settings. The GPIB address, the calibration constants and any protected user data are left untouched.

Factory Options

The instrument's options are held as constants in non-volatile memory. This dialog writes them. Nothing here is in any Tektronix manual; the words, their values and their meanings were read out of the firmware images and confirmed on the bench.

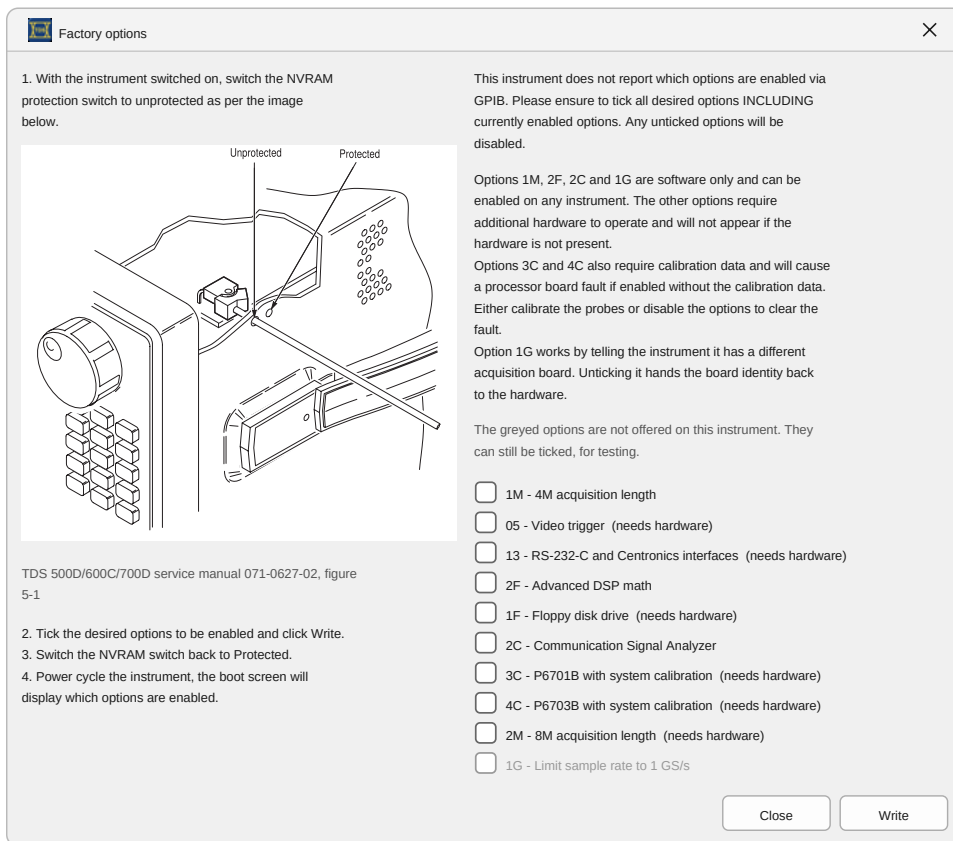


Figure 8-2: The Factory options dialog



WARNING. Enabling an option the instrument's hardware cannot support does not add the hardware. Options 3C and 4C raise a processor board fault if they are enabled without their calibration data; either calibrate the probes or disable the options again to clear the fault.

The Procedure

1. *Unprotect the memory:* With the instrument switched on, set the NVRAM protection switch on the side of the cabinet to unprotected. See Figure 8–3.
2. *Choose the options:* Tick every option that is to be enabled, including the ones already enabled. Anything left unticked is disabled.
3. *Write them:* Press **Write** and confirm.
4. *Protect the memory again:* Set the switch back to protected.
5. *Restart the instrument:* Power cycle it. The boot screen lists the options that are enabled.

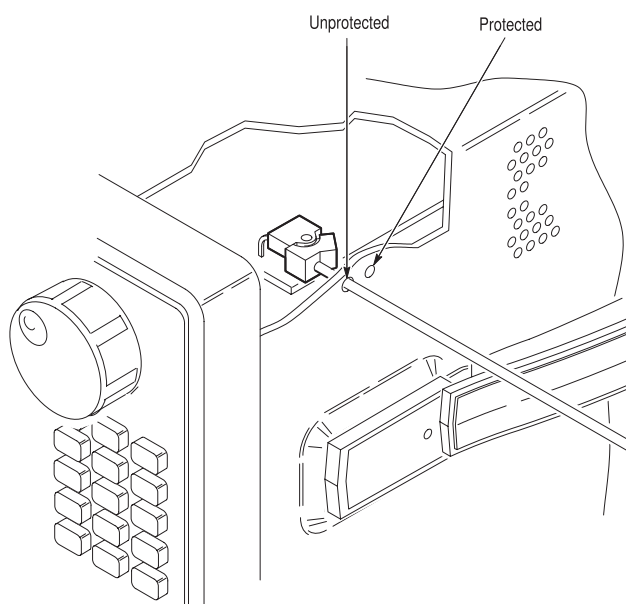


Figure 8–3: Accessing the protection switch, from Tektronix service manual 071-0627-02



CAUTION. With the switch protected, the writes are accepted and nothing changes. Nothing can be read back to confirm what was written: the instrument does not report its option words, and ID? reports only what the firmware believes. The boot screen is the only report.

The Option Words

Every word is written on every press, ticked or not, with the value below when ticked and 0 when not.

Table 8–2: The factory option words

Code	Option	Word	On
1M	4M acquisition length	327686	1
05	Video trigger	327687	1
13	RS-232-C and Centronics interfaces	327688	1
2F	Advanced DSP math	327689	1
1F	Floppy disk drive	327690	1
2C	Communication Signal Analyzer	327692	1
3C	P6701B with system calibration	327693	1
4C	P6703B with system calibration	327694	1
2M	8M acquisition length	327695	1
1G	Limit sample rate to 1 GS/s	131219	14

Options 1M, 2F, 2C and 1G are software only and take effect on any instrument whose firmware reads them. The others need hardware that has to be there.

NOTE. *1G is not an option word at all. The firmware reports 1G when the acquisition board identifies itself as board 14; word 131219 is the constant that overrides that identity when it is not zero. Writing 14 tells the instrument it has a different acquisition board; writing 0 hands the identity back to the hardware.*

An option greyed in the dialog is one this instrument is not offered. For every code but 1G that means the firmware has no getter for it and will never report it however its word is set; Appendix A lists what each firmware reads. 1G is offered only on a TDS 540B, which is the one instrument it has been traced on, although other firmware reads the constant. A greyed option can still be ticked and written, which is how you find out whether a particular instrument takes it.

PASSWORD PITBULL

*OPC?

WORDCONSTANT:ATPUT 327686, 1 (and so on, all ten)

*OPC? (after each)

What the Program Sends

The password is what unlocks the constant-writing commands. Each write is followed by an operation-complete query, and any event the instrument raises is collected and reported as a refusal.

Settings

The Settings tab holds what belongs to this program rather than to the instrument: the colours everything is drawn in, the size of a saved picture, and where the settings and the log are kept.

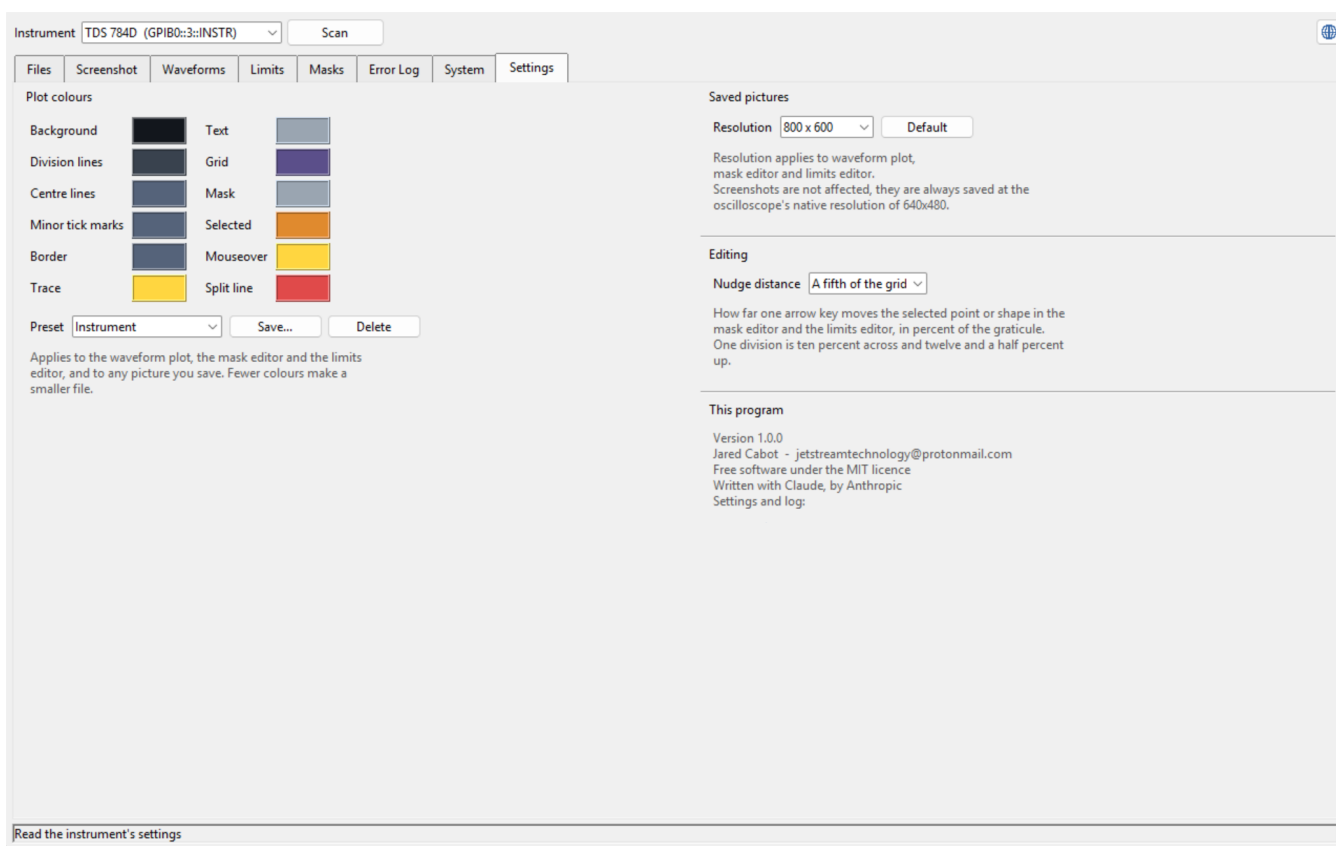


Figure 9–1: The Settings tab

Plot Colours

Twelve colours are used by the waveform plot, the mask editor, the limits editor and every picture saved from them. Click a swatch to change it. There is nothing to apply and nothing to cancel: a colour takes effect as it is picked, which is the only way to judge one.

Table 9–1: What each colour draws

Colour	Where it is used
Background	Behind the graticule
Division lines	The graticule's own lines
Centre lines	The heavier cross through the centre
Minor tick marks	The small ticks along the centre cross
Border	The graticule's outer border
Trace	A captured trace, where the scheme is not the instrument's own
Text	Captions, markers and readouts
Grid	The editor's drawing grid, and the template band
Mask	Mask and limit shapes
Selected	A point or shape that is selected
Mouseover	Whatever the pointer is over
Split line	The line the scissors would cut along

Presets

Four schemes are supplied:

Instrument	The instrument's own screen colours, dark. Under this scheme a captured trace is drawn in the colour the instrument itself draws that channel in.
Printed page	Dark on white, for printing and for pasting into a document.
Phosphor	Green on near-black, the way a storage tube looked.
High contrast	White on black, with no half tones.

Save... keeps the colours you have chosen under a name of your own, and **Delete** removes one you have saved. The four supplied schemes cannot be deleted, but saving one of the same name replaces it.

Saved Pictures

The **Resolution** box sets the size of a picture saved from the waveform plot, the mask editor or the limits editor. The sizes offered are 640 x 480, 800 x 600, 1024 x 768, 1280 x 960, 1600 x 1200 and 1920 x 1440; the default is 800 x 600. All are in the instrument's own four to three, because the graticule is, and the graticule is centred in whatever it is given rather than stretched to fill it.

NOTE. *This does not apply to screenshots. A screenshot is the instrument's own picture and is always saved at its native resolution of 640 x 480.*

Editing

Nudge distance sets how far one arrow key moves the selected point or shape in the mask editor and the limits editor. It is a percentage of the graticule and is the same both ways, so it is worth remembering that a division is ten percent across and twelve and a half percent up.

The first entry, **A fifth of the grid**, is the default and follows whatever the editor's grid spacing is, so five presses cross one square and the keys and the snapping agree with one another. Choosing a distance instead fixes it whatever the grid is doing.

Language

The globe button at the right-hand end of the bar above the tabs lists every language file found beside the program. The choice is remembered. With none remembered the program follows the language Windows is set to, and falls back to English.

Nine catalogues are supplied. A language can be added or corrected without rebuilding anything: a `lang` folder beside the program overrides the one inside it. Copy `en.json`, name it for the language tag, fill in the `_meta` block and translate the values, leaving the keys alone.

NOTE. *The keys are the English text, so anything left out falls back to English on its own and a half-finished file is useful straight away. Placeholders such as `%s` and `%d` must survive translation in the same number and, where they are not named, in the same order. Run the program with `--check-translations` to audit every file before using it.*

Where Things Are Kept

The bottom box names the folder the program keeps its files in, which is the folder the program itself is in.

tdstoolkit.json	The settings. Written as they are changed. A file that will not parse is moved aside to tdstoolkit.json.bad and the program carries on with defaults.
tdstoolkit.log	What the program did, what the instrument reported that was not expected, and any internal fault. Moved to tdstoolkit.log.old at 512 kbytes; one generation is kept.
masks	The mask library.
capabilities.json	What is known about each instrument's firmware. A file here overrides the one inside the program, so an instrument can be added without rebuilding anything.

NOTE. *Settings are not saved if the folder cannot be written to. A read-only folder is a good reason not to save and a poor reason to interrupt what you were doing, so the program says nothing and carries on.*

Firmware Capability

What an instrument can do is decided by its firmware. The table below was read out of the firmware images themselves: the SCPI keyword table decoded from the ROM says directly whether a command exists, and the option words say which option codes the firmware reads. The method was validated against three instruments checked on the bench and reproduces all three exactly.

The program consults this table when it connects. An instrument that is not in it is asked instead, and works without anyone doing anything; an entry that turns out to be wrong is corrected against the instrument and written to the log. The table is `capabilities.json`, and a copy beside the program overrides the one inside it.

Files and Transfers

Read is the command a file's contents can be fetched with: `READFILE` where the firmware has it, `PRINT` where it does not, and none at all where neither exists.

Write is whether `FILESYSTEM:WRITEFILE` exists.

Table A-1: Filesystem and transfer commands by firmware

Model	Firmware	Filesystem	Read	Write
TDS 520	v2.16e	no	none	no
TDS 520A	v3.8.7e	yes	PRINT	no
TDS 520B	v4.1e	yes	PRINT	no
TDS 520C	v5.0e	yes	READFILE	yes
TDS 520C	v5.1.1e	yes	READFILE	yes
TDS 520C	v5.2e	yes	READFILE	yes
TDS 520C	v5.3e	yes	READFILE	yes
TDS 524A	v3.8.2e	yes	PRINT	no
TDS 524A	v3.8.7e	yes	PRINT	no
TDS 540	v2.16e	no	none	no
TDS 540A	v3.8.3e	yes	PRINT	no
TDS 540B	v4.1e	yes	PRINT	no
TDS 540C	v5.0e	yes	READFILE	yes
TDS 540C	v5.1.1e	yes	READFILE	yes
TDS 540C	v5.2e	yes	READFILE	yes
TDS 540C	v5.3e	yes	READFILE	yes
TDS 544A	v3.8.4e	yes	PRINT	no
TDS 580C	v5.0e	yes	READFILE	yes
TDS 580C	v5.1.1e	yes	READFILE	yes
TDS 580C	v5.2e	yes	READFILE	yes
TDS 580C	v5.3e	yes	READFILE	yes
TDS 620	v2.04e	no	none	no

Table A-2: Filesystem and transfer commands by firmware (cont.)

Model	Firmware	Filesystem	Read	Write
TDS 620A	v3.8.5e	yes	PRINT	no
TDS 620B	v4.2e	yes	PRINT	no
TDS 620B	v4.4.1e	yes	PRINT	no
TDS 640A	v3.8.8e	yes	PRINT	no
TDS 644A	v3.8.4e	yes	PRINT	no
TDS 644A	v3.8.5e	yes	PRINT	no
TDS 644B	v4.2e	yes	PRINT	no
TDS 680B	v4.2e	yes	PRINT	no
TDS 680B	v4.4.1e	yes	PRINT	no
TDS 684B	v4.4.1e	yes	PRINT	no
TDS 684C	v5.1e	yes	READFILE	yes
TDS 714L	v7.1.1e	yes	READFILE	yes
TDS 714L	v7.2e	yes	READFILE	yes
TDS 714L	v7.4e	yes	READFILE	yes
TDS 744A	v1.0.1e	yes	PRINT	no
TDS 744A	v1.0e	yes	PRINT	no
TDS 744A	v1.1e	yes	PRINT	no
TDS 744A	v4.0e	yes	PRINT	no
TDS 744A	v4.2.1e	yes	PRINT	no
TDS 754C	v5.0e	yes	READFILE	yes
TDS 754C	v5.2e	yes	READFILE	yes
TDS 754C	v5.3e	yes	READFILE	yes

Table A-3: Filesystem and transfer commands by firmware (cont.)

Model	Firmware	Filesystem	Read	Write
TDS 754D	v6.0e	yes	READFILE	yes
TDS 754D	v6.3e	yes	READFILE	yes
TDS 754D	v6.4e	yes	READFILE	yes
TDS 754D	v6.6e	yes	READFILE	yes
TDS 754D	v7.1.1e	yes	READFILE	yes
TDS 754D	v7.2e	yes	READFILE	yes
TDS 754D	v7.4e	yes	READFILE	yes
TDS 784C	v5.0e	yes	READFILE	yes
TDS 784C	v5.2e	yes	READFILE	yes
TDS 784C	v5.3e	yes	READFILE	yes
TDS 784D	v6.0e	yes	READFILE	yes
TDS 784D	v6.3e	yes	READFILE	yes
TDS 784D	v6.4e	yes	READFILE	yes
TDS 784D	v6.6e	yes	READFILE	yes
TDS 784D	v7.1.1e	yes	READFILE	yes
TDS 784D	v7.2e	yes	READFILE	yes
TDS 784D	v7.4e	yes	READFILE	yes
TDS 794D	v6.0e	yes	READFILE	yes
TDS 794D	v6.3e	yes	READFILE	yes
TDS 794D	v6.4e	yes	READFILE	yes
TDS 794D	v6.6e	yes	READFILE	yes
TDS 794D	v8.0e	yes	READFILE	yes

Factory Options

A tick means that firmware reads that option's word. An option with no tick has no getter in the ROM, so the instrument will never report it however its word is set. This says nothing about whether the hardware the option needs is fitted.

Table A-4: Option codes each firmware reads

Model	Firmware	1M	05	13	2F	1F	2C	3C	4C	2M	1G
TDS 520	v2.16e	X	X	X	X						
TDS 520A	v3.8.7e	X	X	X	X	X					
TDS 520B	v4.1e	X	X	X	X	X					X
TDS 520C	v5.0e	X	X	X	X	X	X	X	X		X
TDS 520C	v5.1.1e	X	X	X	X	X	X	X	X		X
TDS 520C	v5.2e	X	X	X	X	X	X	X	X		X
TDS 520C	v5.3e	X	X	X	X	X	X	X	X		X
TDS 524A	v3.8.2e	X	X	X	X	X					
TDS 524A	v3.8.7e	X	X	X	X	X					
TDS 540	v2.16e	X	X	X	X						
TDS 540A	v3.8.3e	X	X	X	X	X					
TDS 540B	v4.1e	X	X	X	X	X					X
TDS 540C	v5.0e	X	X	X	X	X	X	X	X		X
TDS 540C	v5.1.1e	X	X	X	X	X	X	X	X		X
TDS 540C	v5.2e	X	X	X	X	X	X	X	X		X
TDS 540C	v5.3e	X	X	X	X	X	X	X	X		X
TDS 544A	v3.8.4e	X	X	X	X	X					
TDS 580C	v5.0e	X	X	X	X	X	X	X	X		X
TDS 580C	v5.1.1e	X	X	X	X	X	X	X	X		X
TDS 580C	v5.2e	X	X	X	X	X	X	X	X		X
TDS 580C	v5.3e	X	X	X	X	X	X	X	X		X
TDS 620	v2.04e	X	X	X	X						

Table A-5: Option codes each firmware reads (cont.)

Model	Firmware	1M	05	13	2F	1F	2C	3C	4C	2M	1G
TDS 620A	v3.8.5e	X	X	X	X	X					
TDS 620B	v4.2e	X	X	X	X	X					
TDS 620B	v4.4.1e	X	X	X	X	X					
TDS 640A	v3.8.8e	X	X	X	X	X					
TDS 644A	v3.8.4e	X	X	X	X	X					
TDS 644A	v3.8.5e	X	X	X	X	X					
TDS 644B	v4.2e	X	X	X	X	X					
TDS 680B	v4.2e	X	X	X	X	X					
TDS 680B	v4.4.1e	X	X	X	X	X					
TDS 684B	v4.4.1e	X	X	X	X	X					
TDS 684C	v5.1e	X	X	X	X	X					
TDS 714L	v7.1.1e	X	X	X	X	X	X	X	X	X	X
TDS 714L	v7.2e	X	X	X	X	X	X	X	X	X	X
TDS 714L	v7.4e	X	X	X	X	X	X	X	X	X	X
TDS 744A	v1.0.1e	X	X	X	X	X					
TDS 744A	v1.0e	X	X	X	X	X					X
TDS 744A	v1.1e	X	X	X	X	X					X
TDS 744A	v4.0e	X	X	X	X	X					X
TDS 744A	v4.2.1e	X	X	X	X	X					X
TDS 754C	v5.0e	X	X	X	X	X	X	X	X		X
TDS 754C	v5.2e	X	X	X	X	X	X	X	X		X
TDS 754C	v5.3e	X	X	X	X	X	X	X	X		X

Table A-6: Option codes each firmware reads (cont.)

Model	Firmware	1M	05	13	2F	1F	2C	3C	4C	2M	1G
TDS 754D	v6.0e	X	X	X	X	X	X	X	X		X
TDS 754D	v6.3e	X	X	X	X	X	X	X	X	X	X
TDS 754D	v6.4e	X	X	X	X	X	X	X	X	X	X
TDS 754D	v6.6e	X	X	X	X	X	X	X	X	X	X
TDS 754D	v7.1.1e	X	X	X	X	X	X	X	X	X	X
TDS 754D	v7.2e	X	X	X	X	X	X	X	X	X	X
TDS 754D	v7.4e	X	X	X	X	X	X	X	X	X	X
TDS 784C	v5.0e	X	X	X	X	X	X	X	X		X
TDS 784C	v5.2e	X	X	X	X	X	X	X	X		X
TDS 784C	v5.3e	X	X	X	X	X	X	X	X		X
TDS 784D	v6.0e	X	X	X	X	X	X	X	X		X
TDS 784D	v6.3e	X	X	X	X	X	X	X	X	X	X
TDS 784D	v6.4e	X	X	X	X	X	X	X	X	X	X
TDS 784D	v6.6e	X	X	X	X	X	X	X	X	X	X
TDS 784D	v7.1.1e	X	X	X	X	X	X	X	X	X	X
TDS 784D	v7.2e	X	X	X	X	X	X	X	X	X	X
TDS 784D	v7.4e	X	X	X	X	X	X	X	X	X	X
TDS 794D	v6.0e	X	X	X	X	X	X	X	X		X
TDS 794D	v6.3e	X	X	X	X	X	X	X	X	X	X
TDS 794D	v6.4e	X	X	X	X	X	X	X	X	X	X
TDS 794D	v6.6e	X	X	X	X	X	X	X	X	X	X
TDS 794D	v8.0e	X	X	X	X	X	X	X	X	X	X

Three things are worth drawing out. Option 2M is read only by v6.3e and later, so the 754D, 784D and 794D at v6.0e do not have it. Option 1G is on the 520B and 540B and most 744A versions, but not the 620B, 644B, 680B or 684B. And the 684C at v5.1e is the only C or D image with none of 2C, 3C, 4C or 1G.

File Formats

Waveforms

The formats this program reads and writes, and what each one keeps.

Table B-1: Waveform formats

Extension	Read	Write	What it holds
.isf	yes	yes	The instrument's own transfer format: the WFMPRE preamble and the curve exactly as CURVE? returned it. Byte for byte on a round trip
.csv	yes	yes	A units header, then scaled x,y pairs. Re-quantised when read back, to within one step of the eight-bit range
.tdw	yes	yes	The instrument's own layout under this program's own extension, so a file it wrote is never mistaken for one an instrument wrote. Sixteen-bit samples
.wfm	yes	no	A file an instrument wrote, which it can recall into a reference. Written only onto the instrument's own disk, by Send to instrument
.png	no	yes	The graticule as it stands, with the record strip, at the resolution set on the Settings tab

NOTE. A waveform file from a TDS5000, TDS6000 or TDS7000, or from a DPO or MSO, is a different and much later format. The program recognises one and says so rather than misreading it.

What a .WFM Carries

The instrument's own file holds more than the samples. All of the following is read out of it and written back into it:

Table B-2: The fields in a .WFM

Field	What it is
Record	The samples, sixteen bits each, with the sixteen acquired points either side of the record that the instrument uses to interpolate the display
Timebase	Seconds a sample, and the duration of the whole record
Trigger	Its position, as a whole percent of the record
Vertical	Volts a division, the offset in divisions, and the voltage at that offset
Units	Horizontal and vertical, separately
Source	Which input it came from: a channel, a maths waveform or a reference
Coupling	DC, AC or GND
Acquisition	Sample, Average, Envelope, Hi Res or Peak Detect
Format	Y or ENV

There is no XZERO and no date. The file holds the record and how to scale it, not when it was taken.

Masks and Setups**Table B-3: Mask and setup formats**

Extension	What it holds
.MSK	A mask, in either this program's text format or the Tektronix i-Pattern binary format. Which one a file is is established from its contents, and saving keeps the format it came in as
.SET	The instrument settings a mask was drawn against, as SCPI. Sits beside the mask under the same name. Two REM lines are written and read back by this program: the unit interval, and whether the signal is differential

Table B–4: Mask limits

Quantity	Value
Segments a mask	8
Points on segment	50
Minimum points on segment	2
Coordinates	0 to 100 percent, X across, Y up
Masks in an i-Pattern file	10 by the count field; 9 fit the 2023-byte file

Screenshots

A screenshot is saved either in the format the instrument sent or as PNG, and either way it carries any rotation or inversion applied here. Untouched, the native file is the instrument's own bytes exactly. The suffix offered for it is taken from those bytes, not from the format that was asked for.

The Program's Own Files

Table B–5: Files beside the program

File	What it is
tdstoolkit.json	Settings: language, address, colours, saved colour schemes, picture resolution, nudge distance
tdstoolkit.log	What was done and what the instrument reported. Rotated at 512 kbytes
capabilities.json	The firmware capability table of Appendix A
masks	The mask library
lang	The language catalogues

If Something Goes Wrong

Connecting

Table C-1: The instrument does not answer

What you see	What to check
The picker opens by itself at start-up	Nothing answered at the remembered address. Press Scan in the picker; the instrument may have moved or be switched off
The scan finds nothing at all	The GPIB driver and the cable. VISA lists addresses it has been configured with whether or not anything is there, so an empty result means nothing on the bus answered
The scan finds other instruments but no oscilloscope	That the oscilloscope is in Talk/Listen, not Off Bus and not Hardcopy (Talk Only). In Talk Only it will not answer a query
An address answers as in use by another program	Another program holding the VISA session. Only one program at a time can have it

Transfers

Table C-2: Files and transfers

What you see	What it means
Upload is greyed for the whole session	This firmware has no FILESYSTEM:WRITEFILE. Every A and B series instrument is in this position. Nothing can send it a file over GPIB
Download is greyed for the whole session	This firmware has neither READFILE nor PRINT
"Nothing to browse" when connecting	Firmware v2.x, which has no filesystem commands at all. The Waveforms tab still works
"No disk in the drive"	The floppy drive was listed and answered, and holds no disk
An upload reports that it was not verified	The file was written and read back and the two did not match. Nothing is assumed: try again, and check the instrument's own screen

Waveforms and Screens

Table C-3: Capture

What you see	What it means
"Waveform capture is not supported in DPO mode"	In DPO the instrument fills a pixel database instead of a waveform record, and every point reads 128. Take a screenshot, or leave DPO
A source cannot be read	A channel that is not displayed cannot be read; the instrument answers event 2241. Choosing a greyed name switches it on
"This instrument offers no image format this program can read"	Every hardcopy format this firmware offers is a printer language rather than an image
A capture times out or the next command does	A hardcopy started too soon after the previous one is ignored, and a half-sent one blocks the bus. The program issues a device clear after a failure; give it a moment before trying again

Masks and Limits

Table C-4: Testing

What you see	What it means
The instrument draws the mask differently	It stores the points exactly and joins them its own way. Three arrangements are redrawn; the program names them before sending. Cutting the shape into simpler pieces avoids it
Send and Start measurement greyed for the whole session	This firmware has no Option 2C, so it has no mask subsystem to send to. Drawing, saving and loading still work; use the Limits tab to judge a signal
An eye that will not centre	Check that the sweep is holding about one bit. Send the setup beside the mask, which names the timebase the mask was drawn against
A limit test that passes everything	Check that the template was learnt from the same channel at the same volts a division. A template built against another channel's scaling lands in the wrong place

The Log

Anything unexpected goes to `tdstoolkit.log` beside the program, with the time and what was being done. An event the instrument raised that the program did not expect is shown once, written to the log, and put on the status line every time it happens afterwards. A session in which nothing goes wrong writes nothing at all.

A fault is written with a few lines in front of it saying what the program was connected to at the time: the instrument, the transfer commands its firmware has, the options fitted and the sources it was displaying. They are the first things anyone reading the fault needs, and by then it is too late to ask.

An internal fault in the program is caught, written to the log with its traceback, and reported without closing the window. Nothing is sent to the instrument in that case.